When Parents Can Spare Time: Paid Sick Leave and Adolescent Mental Health*

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Abstract

Adolescent mental health care often depends on parental time: parents must recognize need, schedule appointments, provide consent and transportation, and often attend visits. We study whether state paid sick leave (PSL) mandates affect adolescents' mental health care and self-reported symptoms, combining the National Survey of Children's Health (2016–2024) and the Youth Risk Behavior Surveillance System (2005–2023) with a staggered difference-in-differences design exploiting variation in state PSL adoption timing. PSL mandates increase receipt of any mental or behavioral health care in the past year by 3.0 percentage points, a 16.4 percent increase over the untreated baseline; medication use and parent-reported diagnoses also rise. In the independent YRBSS data, depressive symptoms decline, while suicidal ideation and suicide planning are negative but less precisely estimated. This pattern of higher care receipt and lower self-reported symptoms is more consistent with improved detection and treatment initiation than with worsening underlying mental health. PSL also increases parental care-coordination time by 3.2 percentage points, with no reduction in work hours or employment. Short, protected leave can help parents initiate adolescent mental health care without measurable labor-market costs.

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1 Introduction

Children and adolescents in the United States face a growing mental health crisis, with rising rates of depression, anxiety, and suicidality over the past decade (Centers for Disease Control and Prevention, 2024; Xiang et al., 2024; Warnick and Kolade, 2024). Unmet mental health needs among youth remain widespread, especially for conditions such as depression and behavioral disorders that are often underdiagnosed and untreated until symptoms escalate (Lu, 2019; The Pew Charitable Trusts, 2024). These gaps have long-term consequences for educational attainment, employment, and health over the life course (Bubonya et al., 2017; Kondirolli and Sunder, 2022; Frijters et al., 2014; Bryan et al., 2022; Fletcher, 2013; Currie, 2024). A central but often overlooked access input is parental time. For adolescents, mental health care typically requires parents to schedule appointments, coordinate insurance, provide consent, arrange transportation, and attend visits that often occur during standard working hours. Limited parental time flexibility can therefore delay treatment initiation and reduce adherence (Mitchell and Selmes, 2007; Hansen et al., 2021).

Paid sick leave (PSL) mandates are a natural policy lever for this margin. PSL laws allow workers to take short, paid, job-protected absences for personal or family health needs without risking income loss or retaliation (Eisenberg et al., 2025; Seixas and Macinko, 2020; Callison et al., 2025). Prior work shows that PSL mandates affect workers' own health care use and generate family spillovers, including changes in parental caregiving time, child maltreatment, and care provision (Chen et al., 2020; Callison and Pesko, 2022; Maclean and Pabilonia, 2025; Deza et al., 2025; Guo and Peng, 2025). Much less is known about whether these labor protections improve adolescent mental health care access, where treatment is often parent-mediated.

The U.S. institutional setting makes this question especially policy-relevant. Unlike most high-income countries, the United States lacks a permanent federal paid sick leave mandate; access depends on state mandates or voluntary employer provision. As of 2024, 18 states and the District of Columbia have enacted statewide PSL mandates that typically provide

roughly 3–7 paid days per year at full wage replacement and explicitly permit leave for family caregiving, including children's medical appointments (Pichler and Ziebarth, 2024; Mitchell, 2024). Employer-provided PSL access is lowest among low-wage, part-time, and service-sector workers (Bureau of Labor Statistics, 2024), so state mandates expand legal entitlement most among families facing the steepest time and financial constraints. Whether that legal entitlement translates into completed adolescent mental health care is an empirical question.

Against this background, this paper provides, to our knowledge, among the first quasi-experimental evidence on whether state PSL mandates affect adolescents' receipt of mental health care and their self-reported symptoms. We link state PSL mandate timing to adolescent-level survey data from two complementary sources: the National Survey of Children's Health (NSCH, 2016–2024) and the Youth Risk Behavior Surveillance System (YRBSS, 2005–2023), exploiting staggered state PSL adoption with the interaction-weighted estimator of Sun and Abraham (2021), which is designed for settings with staggered treatment timing and heterogeneous treatment effects. The NSCH is the primary data source because it directly measures parent-reported receipt of mental or behavioral health care, medication use, and current clinical diagnoses. These outcomes capture engagement with the care system, but they also raise an interpretive challenge: higher care and diagnosis rates could reflect either better detection and treatment of existing need or worsening underlying mental health. The YRBSS addresses this ambiguity by measuring adolescents' own self-reported depressive symptoms and suicidal ideation in an independent survey, with a longer pre-period and a separate respondent.

The results show that PSL mandates increase adolescents' engagement with the mental health care system. In the NSCH, PSL raises the probability that an adolescent received any mental or behavioral health care in the past year by 3.0 percentage points ($p < 0.01$), a 16.4% increase relative to the untreated baseline mean of 18.2%. Medication use rises by 1.7 percentage points, and parent-reported current depression and anxiety diagnoses

rise by 1.6 and 1.3 percentage points, respectively. We interpret the diagnosis outcomes as complementary evidence because diagnosis rates can rise through increased clinical detection and reporting, not only through changes in underlying morbidity. The YRBSS points in the same substantive direction: PSL reduces self-reported depressive symptoms by 2.2 percentage points and suicidal ideation by 0.8 percentage points, relative to untreated baseline means of 31.6% and 16.9%, respectively; the suicide-planning estimate is also negative but imprecise. The joint pattern of higher care receipt, positive diagnosis estimates, and lower self-reported symptoms is more consistent with improved detection and treatment initiation than with deterioration in underlying adolescent mental health.

To probe the channel, we examine parental time reallocation. PSL increases the share of families spending any time coordinating the adolescent's care by 3.2 percentage points, a 28% increase relative to baseline, with no accompanying reduction in work hours or employment. This pattern is consistent with prior evidence that PSL affects time-sensitive health care use and caregiving behavior (Chen et al., 2020; Maclean and Pabilonia, 2025; Guo and Peng, 2025), and points to a short appointment-day channel rather than broader changes in family functioning or labor supply.

The closest existing study is Eisenberg et al. (2025), who study PSL mandates and mental health care use among working-age adults. Their central finding is that PSL does not increase overall adult mental health care utilization, but instead shifts treatment toward more time-intensive care, especially psychotherapy. That result is important for this paper because it shows that PSL can relax time constraints around mental health treatment, but on the adult intensive margin: among adults who can seek care for themselves, protected time changes the type or intensity of care rather than whether any care occurs. Adolescents face a different care-access process. They often cannot independently schedule, consent to, travel to, or navigate insurance requirements for mental health treatment, so parental time constraints can bind before care begins. The adolescent setting therefore points to a different margin of response: PSL operates at the extensive margin of whether care is initiated at all.

We test that dependent-care margin directly by examining whether parental access to protected leave translates into adolescent mental health care receipt, and by pairing parent-reported care and diagnosis outcomes with adolescent self-reported symptoms. This design contributes to three strands of literature. First, it adds to research on the health effects of labor-market policies (Chen et al., 2020; Pichler and Ziebarth, 2017; Callison and Pesko, 2022; Guo, 2025) by showing that PSL can affect dependent populations through a parental-mediation channel. Second, it extends evidence on intergenerational spillovers of family-oriented labor protections (Maclean and Pabilonia, 2025; Deza et al., 2025; Bullinger, 2019) from early childhood and family stability to adolescent clinical engagement. Third, it speaks to a broader policy question about adolescent mental health access: short-duration workplace leave can matter at the point of treatment initiation, where parents often supply the time needed to move from concern to care.

Section 2 reviews the institutional background on U.S. paid sick leave policy. Section 3 describes the data sources. Section 4 presents the empirical strategy. Section 5 presents results on mental health care utilization, diagnoses, and mental health outcomes. Section 6 concludes with a discussion of implications.

2 Institutional Background

2.1 Adolescent Mental Health Care in the United States

Adolescent mental health care in the United States spans multiple treatment settings with varying time requirements. Primary care physicians provide initial mental health screening, brief counseling, and medication initiation (Zuckerbrot et al., 2018; Cheung et al., 2018). More intensive outpatient mental health care typically includes comprehensive diagnostic evaluation, individual psychotherapy, and medication management delivered by specialty mental health providers. Outpatient mental health services are provided by psychiatrists and other mental health professionals and commonly involve psychotherapy and psychotropic

medication use (Olfson et al., 2019; Centers for Medicare & Medicaid Services, 2025). Standard psychotherapy sessions typically last 45–60 minutes (Walkup et al., 2008; Weisz et al., 2017), with evidence-based treatments such as cognitive behavioral therapy generally requiring weekly sessions over 8–20 weeks (March et al., 2007; Weisz et al., 2017). These treatment structures require sustained, repeated parental time commitments across weeks, establishing why time—not only coverage or cost—is an operative constraint on adolescent access.

A defining feature of adolescent mental health care is that, even where minors can consent to some services, access often remains practically and legally mediated by parents. Eighteen states either prohibit independent adolescent consent to mental health treatment or provide no explicit authorization for minors to consent independently (Schleider et al., 2025). Parents must schedule appointments, coordinate with insurance providers, arrange transportation, and often attend family therapy sessions or medication consultations. These logistical requirements create structural dependence on parental time availability that distinguishes adolescent from adult mental health care access. Provider shortages compound these time barriers: approximately 70% of U.S. counties lack a single child or adolescent psychiatrist (McBain et al., 2019), and simulated-patient studies find that fewer than 17% of child psychiatrists have available appointments for new patients (Cama et al., 2017). Consequently, even when families have insurance coverage and motivation to seek care, time constraints—including inflexible work schedules and lengthy wait times—can delay treatment initiation and reduce treatment adherence.

Insurance coverage is a separate but related barrier. The Mental Health Parity and Addiction Equity Act of 2008 and the Affordable Care Act extended parity requirements to most plans (Barry et al., 2010; Mehrotra and Busch, 2024), yet parity alone produced limited gains in mental health treatment receipt (Drake et al., 2019), in part because it addresses financial terms rather than time, provider availability, or information barriers. Coverage among U.S. adolescents is high but varies in type—employer-sponsored, Medicaid/CHIP, marketplace, military, and other—and these types differ in network adequacy and adminis-

trative friction even when nominal coverage is present, so insurance status alone need not determine whether a family can convert protected leave into a completed mental health visit.

School-based mental health services are a parallel access channel that can operate partly outside the parental-time-and-appointment constraint that PSL may ease, though many school-based services still require parental consent, referral, or coordination. Approximately 60% of U.S. public schools provide some form of mental health services on site, including counseling, screening, and case management, though intensity and staffing vary substantially across districts (National Center for Education Statistics, 2022; The White House, 2022). School-based services can substitute partially for clinic-based care for milder cases—particularly initial screening and short-term counseling—but typically do not provide the diagnostic evaluation, psychiatric medication management, or sustained psychotherapy that defines specialty mental health treatment (Ali et al., 2019). These features mean that parental time constraints can remain relevant even when school-based services are available.

These structural features create a framework in which parental time availability functions as a critical enabling factor for treatment access. Cross-sectional evidence is consistent with an association between parental PSL access and children's healthcare utilization, with lower reliance on emergency services (Asfaw and Colopy, 2017; Seixas and Macinko, 2020). Although typical PSL mandates provide roughly 3–7 paid days per year—insufficient to cover a full course of weekly psychotherapy—even a modest expansion in protected time may be consequential at the margin, where the binding constraint is often the inability to attend an initial evaluation or a few early sessions without risking income loss or job security. State paid sick leave mandates therefore represent a policy intervention targeting time-related barriers at the critical margin of treatment initiation.

2.2 Paid Sick Leave Policy in the United States

The United States does not have a permanent federal paid sick leave mandate, making it an outlier among high-income countries. The Family and Medical Leave Act of 1993 provides up

to 12 weeks of unpaid leave for eligible workers and can cover some intermittent medical leave, but it is unpaid, eligibility-limited by firm size, job tenure, and hours worked, and poorly suited to routine short-term appointment needs (Appelbaum, 2014; Pichler and Ziebarth, 2024).

In the absence of federal paid sick leave, policy has developed through state and local legislation. San Francisco enacted the first employer mandate in 2007, followed by the District of Columbia in 2008 and Connecticut in 2012 (Ober, 2006; Gilliam and Ben-Ishai, 2012). As of 2024, 18 states and DC have enacted statewide PSL mandates (Mitchell, 2024). These mandates share common structural features: employees typically accrue paid sick leave at a rate of one hour per 30–40 hours worked, with annual caps of 24–56 hours (3–7 days) compensated at 100% wage replacement (U.S. Department of Labor, 2023; Paycor, Inc., 2025). Critically for this study, leave can be used not only for employees' own health needs but also for family caregiving, with state mandates explicitly covering care for children, including attending medical appointments or providing care during illness. Mandates also include job protection and anti-retaliation provisions and limit employers' ability to require extensive documentation (Pichler and Ziebarth, 2017).

The empirical literature on PSL mandates establishes several relevant patterns. First, mandates increase access to and use of paid sick leave: employer-provided PSL coverage and worker utilization of sick days rise substantially after adoption (Ahn and Yelowitz, 2016; Callison and Pesko, 2022). Second, PSL affects adults' own health care use, shifting utilization toward planned and preventive services such as prescriptions, vaccinations, and cancer screenings, while reducing some potentially avoidable emergency department use (Chen et al., 2020; Ma et al., 2022; Callison et al., 2023; Maclean et al., 2024). Third, PSL can generate spillovers beyond the worker: prior studies find increases in parental caregiving time, changes in care provision, reductions in child maltreatment, and effects on child disability-related outcomes, consistent with protected leave changing family behavior and child well-being (Maclean and Pabilonia, 2025; Guo and Peng, 2025; Deza et al., 2025; As-

samidanov et al., 2025). Existing evidence therefore shows that PSL can affect time-sensitive care and family outcomes, but leaves open whether parental access to protected leave translates into adolescent mental health care initiation.

Access to PSL remains highly stratified: low-wage, part-time, and service-sector workers are substantially less likely to receive employer-provided PSL absent mandates (Bureau of Labor Statistics, 2024). These same populations face elevated barriers to accessing adolescent mental health care (The Pew Charitable Trusts, 2024). Consequently, state PSL mandates disproportionately expand legal coverage among families most likely to newly gain leave and who also face substantial access barriers. Whether this legal entitlement translates into completed adolescent mental health care depends on whether protected time can be combined with provider availability, insurance access, and scheduling capacity.

These institutional features generate several implications for adolescent mental health. First, by providing parents with protected time to attend appointments, PSL mandates may increase detection and diagnosis of mental health conditions that were previously unidentified, leading to increases in parent-reported diagnoses and treatment receipt even if underlying symptom burden remains unchanged. Second, PSL may improve access to care by enabling parents to attend initial consultations, diagnostic evaluations, medication-management visits, or early therapy sessions that are critical for establishing a treatment plan. Even a small number of additional contacts can be clinically meaningful: evidence from stepped-care models and brief interventions suggests that early engagement with mental health services can reduce depressive symptoms and suicidal ideation, especially for adolescents who would otherwise receive no care (Kaczkurkin and Foa, 2015; Keles and Idsoe, 2018; van Ballegooijen et al., 2025). Third, PSL may affect adolescent mental health indirectly through reduced family stress, improved parental mental health, or increased caregiving time (Miller, 2022; Slopen, 2024; Maclean and Pabilonia, 2025).

These channels generate distinct empirical predictions. Increased detection predicts higher rates of parent-reported diagnoses and treatment receipt. Improved access to effective

treatment predicts reductions in adolescent self-reported symptoms and suicidal ideation. Observing both patterns simultaneously—increased formal care-system contact alongside reduced self-reported symptoms—would support an interpretation in which PSL eases time-related access barriers and improves detection and treatment initiation, rather than simply increasing recorded morbidity.

3 Data

3.1 Paid Sick Leave Policy Data

We begin by constructing a state–year panel of Paid Sick Leave (PSL) mandates, which serves as the primary policy variable in this analysis. Policy data are drawn from the National Partnership for Women and Families' compilation of Paid Sick Days Statutes, supplemented with legislative texts and state labor department releases to verify effective dates and coverage provisions. Each law specifies key implementation features, including accrual rates (typically one hour of paid leave for every 30–40 hours worked), annual caps ranging from 24 to 56 hours (3–7 days per year), and eligible uses such as personal illness, family caregiving, or school closures. We code a binary indicator for whether a state had an active PSL mandate in effect during a given survey year.[1] The final policy panel includes the 18 states and the District of Columbia that adopted statewide PSL mandates by 2024, with adoption dates spanning the District of Columbia's November 2008 law, coded as the 2009 cohort under our post-July rule, through Minnesota's 2024 implementation.

[1]Because NSCH and YRBSS collect information throughout the calendar year, states with laws taking effect after July were coded as treated beginning in the following year to ensure consistent alignment between policy exposure and the survey reference period.

3.2 National Survey of Children's Health (NSCH) Data and Outcomes

Our primary data source is the National Survey of Children's Health (NSCH), a nationally representative cross-sectional survey administered annually by the U.S. Census Bureau on behalf of the Health Resources and Services Administration's Maternal and Child Health Bureau. The NSCH collects detailed information on children's physical, mental, and behavioral health, health care access, family environment, and social determinants of well-being. The survey employs a stratified address-based sampling design and is completed online or by mail by a parent or guardian who is knowledgeable about the child's health and development. Each year, the NSCH includes approximately 35,000 to 45,000 completed child-level interviews that together represent the noninstitutionalized population of children aged 0–17 years in all 50 states and the District of Columbia (U.S. Census Bureau, 2023).

Our analysis uses pooled data from the 2016–2024 NSCH waves, corresponding to the period following the NSCH redesign that standardized sampling procedures, weighting, and questionnaire content across years. We begin the study period in 2016 to ensure consistency in outcome measurement and survey design and extend it through 2024, the newest wave in the updated analytic file. The analytic sample is restricted to adolescents aged 12–17, the age range for which mental health service use, medication use, and diagnostic indicators are most reliably and consistently reported in the NSCH. After applying the age restriction and excluding pre-panel adopter cohorts, the descriptive analytic sample includes 128,231 adolescent observations. The regression sample for the primary outcome, any mental health care, is 123,372 observations after listwise deletion of missing covariates; sample sizes vary slightly across outcomes and are reported in each table.

The unit of analysis is an individual adolescent observed in a given survey year. We use publicly available state identifiers to merge adolescent-level outcomes from the NSCH with a state–year panel capturing the timing of paid sick leave (PSL) mandate adoption and other time-varying state policy and economic conditions. Our primary NSCH outcome is a

binary indicator equal to one if an adolescent received any mental or behavioral health care in the past 12 months, and zero otherwise. We also examine medication use for emotional or behavioral conditions and parent-reported current diagnoses of depression and anxiety. Together, these measures capture distinct stages of the mental health care pathway, ranging from realized service use to pharmacological treatment and diagnostic recognition. Because the NSCH does not identify whether mental health care occurred in school-based or clinic-based settings, we cannot separately estimate effects by service setting.

The NSCH diagnosis variables capture parent-reported clinical diagnoses made by a health professional rather than symptom scales. NSCH separately records whether a child has ever been told by a provider that they have depression or anxiety and whether the child currently has the condition. We use the current-condition indicator throughout: the diagnosis variable equals one if the parent reports that the adolescent currently has the condition, and zero otherwise, including cases where the adolescent was previously diagnosed but no longer currently has the condition. Estimated PSL effects on this measure therefore reflect changes in current clinical identification, not the cumulative stock of ever-received diagnoses.

Table 1 presents full-sample summary statistics for the analytic sample pooled across survey years and separately for states with and without a PSL mandate in effect during the survey year. In the full sample, 15.7% of adolescents received any mental health care in the past 12 months, 12.8% used medication for emotional or behavioral conditions, and 10.0% and 16.4% had parent-reported current diagnoses of depression and anxiety, respectively. Adolescents residing in PSL mandate states are descriptively more likely to receive any mental health care, 17.8% versus 15.4%, and to have anxiety diagnoses, 19.3% versus 15.9%, compared with adolescents in non-mandate states.

The demographic composition of adolescents also differs across states with and without PSL mandates. Adolescents in mandate states are more likely to reside in households with higher parental educational attainment, with 52.6% having a parent with a college degree or higher compared with 46.6% in non-mandate states. Racial and ethnic composition varies

across groups, with PSL mandate states having a lower share of White, non-Hispanic adolescents and a higher share of Hispanic adolescents. Insurance coverage is high in both groups, though slightly higher in mandate states, 92.5% versus 91.5%. At the state level, PSL mandate states tend to exhibit higher minimum wages, broader Medicaid eligibility thresholds for both parents and non-parents, higher housing prices, and greater primary care physician availability, alongside more stringent COVID-19 policy responses. These patterns underscore systematic differences in policy and economic environments across states that motivate the inclusion of rich state-level controls in the regression analysis.

(Table 1 here)

3.3 Youth Risk Behavior Surveillance System (YRBSS) Data and Outcomes

The data on self-reported depression and suicidal ideation among youth are taken from the Youth Risk Behavior Surveillance System (YRBSS), a biennial survey conducted by the U.S. Centers for Disease Control and Prevention (CDC) since 1991. The YRBSS High School Survey is administered to students in grades 9–12, typically ages 14–18, and is designed to be representative at the state level in participating states for monitoring youth health, safety, and risk behaviors. The YRBSS employs a two-stage cluster sampling design, sampling schools in the first stage and classrooms within schools in the second stage. Survey weights are provided to account for differential selection probabilities and nonresponse. We pool ten biennial YRBSS waves from 2005 through 2023, covering both the pre- and post-policy periods relevant for our analysis. Because the YRBSS samples enrolled high-school students, our estimates capture the effect of PSL on the in-school adolescent population. We cannot rule out that PSL adoption affects school attendance, dropout, or chronic absenteeism in ways that shift the composition of sampled students; to the extent the most affected adolescents are also more likely to be out of school, this would tend to attenuate our YRBSS estimates.

We use three distinct measures of adolescent mental health derived from YRBSS questions. The first is self-reported depressive symptoms, captured by the youth response to the question: "During the past 12 months, did you ever feel so sad or hopeless almost every day for two weeks or more in a row that you stopped doing some usual activities?" The question aligns closely with established diagnostic criteria for a depressive episode and is widely used as a measure of depressive symptoms in population-based research. The second is suicidal ideation, which measures whether the respondent seriously considered attempting suicide in the past twelve months. The third is suicide planning, measured by a binary indicator for whether the respondent reports having made a suicide plan in the past 12 months. Appendix Table A1 summarizes these outcomes. In the pooled sample, approximately 30% of students report depressive symptoms, 16.8% report suicidal ideation, and 13.7% report having made a suicide plan; untreated-state means reported with the regression estimates differ slightly. In contrast to the NSCH, YRBSS outcomes capture adolescents' self-reported emotional states and behaviors rather than clinical diagnoses.

These three mental health measures were selected because the phrasing of these questions has remained consistent across all survey waves dating back to 2005. This consistency is critical for our research design, as it provides a pre-intervention period prior to the implementation of Connecticut's state-level Paid Sick Leave (PSL) mandate in 2012. The longer YRBSS panel therefore retains early-adopting states that the NSCH analysis must drop for lack of pre-treatment observations. Although additional mental health items were introduced in more recent YRBSS surveys, we limit our analysis to these three outcomes because they are the only ones with sufficient pre-policy coverage. In addition to these outcome variables, we incorporate individual-level covariates from the survey—specifically age, sex, and race/ethnicity—which we also use to examine heterogeneous treatment effects.

4 Empirical Strategy

We estimate the causal effect of state paid sick leave (PSL) mandates on mental health outcomes using variation in the timing of PSL adoption across U.S. states. States adopt mandates in different years, and once adopted, mandates remain in effect. This staggered rollout creates a difference-in-differences design with multiple treatment cohorts. In settings with staggered treatment adoption, conventional two-way fixed effects (TWFE) estimators can be problematic. When treatment effects are heterogeneous across cohorts or evolve over time, TWFE implicitly compares early-treated units to later-treated units, using already-treated observations as controls. These comparisons can receive negative or non-convex weights, leading to estimates that are difficult to interpret and potentially biased relative to any meaningful average treatment effect (Goodman-Bacon, 2021). Therefore, we use the interaction-weighted event-study estimator of Sun and Abraham (2021), which is robust to staggered adoption and heterogeneous treatment effects. Sun–Abraham is one of a family of recently developed staggered-DiD estimators that address heterogeneous treatment effects, alongside Callaway and Sant'Anna (2021), de Chaisemartin and D'Haultfœuille (2020), and Borusyak et al. (2024); we adopt Sun–Abraham because its interaction-weighted aggregation aligns naturally with our cohort structure and yields interpretable cohort-specific dynamic effects. The Sun–Abraham estimator constructs treatment effects by comparing each treated cohort only to states that are never treated over the sample period, rather than to other treated or not-yet-treated states. The coefficients θ_{gk} represent cohort- and event-time–specific average treatment effects, measuring the effect of PSL adoption k years relative to implementation for states first treated in year g. Aggregate treatment effects are then weighted averages of these cohort-specific estimates, with weights that are non-negative and sum to one, so both the static and event-study coefficients are interpretable as averages of well-defined cohort-level treatment effects.

$$Y_{ist} = \alpha_s + \lambda_t + X'_{st}\beta + I'_{ist}\omega + \sum_{g \in \mathcal{G}} \sum_{\substack{k \in \mathcal{K} \\ k \neq -1}} \theta_{gk} \, \mathbf{1}\{G_s = g\}\mathbf{1}\{t - g = k\} + \varepsilon_{ist}. \tag{1}$$

Equation (1) states the interaction-weighted specification we estimate. Here, Y_{ist} is the outcome for individual i in state s and survey year t; G_s denotes the first treated survey year in state s, with $G_s = \infty$ for never-treated states. Event time is defined as $k \equiv t - g$, and $\mathcal{G}$ denotes the set of treatment years while $\mathcal{K}$ denotes the set of event times retained in the specification, with $k = -1$ omitted as the reference period. For the YRBSS specifications, t indexes biennial survey-wave years rather than annual calendar years. Because YRBSS is observed biennially, the event-study reference period is defined as the last pre-treatment survey wave rather than a literal one-calendar-year lead. This aligns treated cohorts to the closest clean pre-policy observation available in the survey. The specification includes state fixed effects α_s, year fixed effects λ_t, state-level time-varying controls X_{st}, and individual-level controls I_{ist} that vary across the NSCH and YRBSS.

In the NSCH, individual-level controls include adolescent age, sex, race/ethnicity, parental education, number of adults and children in the household, and parental marital status. Household income and insurance type are excluded from the control set because they may themselves respond to PSL adoption. We use them only to define descriptive subgroups, and the balance checks in Appendix Table A9 find that mandate timing does not predict either composition. In the YRBSS, individual-level controls include age, sex, and race/ethnicity. State-level controls include the state minimum wage, unemployment rate, Medicaid income eligibility thresholds for parents and non-parents, community health center and primary care physician supply per capita, the Housing Price Index, log per capita income, COVID-19 case rates, COVID-19 death rates, the Oxford COVID-19 Government Response Index, and concurrent-policy indicators for an active state Paid Family and Medical Leave (PFML) program, the post-2014 expansion of Medicaid reimbursement for school-based health services (Bullinger et al., 2025), and active PTO mandates in states whose laws provide general

paid time off rather than paid sick leave.[2] We estimate Equation (1) using survey sampling weights and cluster standard errors at the state level. Never-treated states serve as the comparison group. The Sun–Abraham estimator recovers cohort-specific dynamic treatment effects and aggregates them using non-negative weights, avoiding contamination from comparisons across differently treated cohorts.

4.1 Identification and Treatment Timing

A key feature of the NSCH design is that the redesigned survey begins in 2016, after several states had already adopted PSL mandates. Table 2 therefore lays out the identifying cohort structure explicitly. Within the 2016–2024 NSCH window, states adopt PSL across six statutory cohorts: Vermont (2017), Arizona/Maryland/Washington (2018), New Jersey/Rhode Island (2019), Colorado/New York (2021), New Mexico (2023), and Minnesota (2024). Because PSL uptake takes time—workers must accrue leave, learn of the benefit, and then schedule and attend appointments—our static specification codes treatment with a one-year lag relative to statutory adoption. The event study, which measures event time relative to statutory adoption, confirms this timing: the care response is essentially zero in the adoption year and emerges the following year (Figure 2). Under this lag, Minnesota's coded treatment year (2025) falls outside the panel, so it contributes pre-treatment support but no post-treatment observations to the static estimate. States whose mandates predate the NSCH panel—Connecticut, the District of Columbia, California, Massachusetts, and Oregon—are excluded from the NSCH Sun–Abraham analysis because they have no pre-treatment observations in the NSCH window. The YRBSS panel begins in 2005 and therefore supplies substantially longer pre-treatment histories for these early-adopting states, which is why we use it as complementary symptom-side evidence rather than as a second primary care-utilization dataset.

[2]Because the YRBSS is administered biennially, state-level controls in the YRBSS specifications are averaged over the two-year survey period to align with the survey reference window.

(Table 2 here)

This cohort structure shapes both estimation and interpretation. The central identifying variation comes from cohorts with multiple pre-treatment years and meaningful post-treatment exposure. We therefore present two NSCH event-study windows. The full-window event study uses all available treated cohorts and is the main dynamic display, binning the deepest lead and the longest lag and showing exact event times in between, with the year before adoption omitted as the reference. The balanced-window event study is an appendix diagnostic that retains only cohorts observed for at least three pre-treatment years, which drops the most thinly supported leads at the cost of a smaller sample. This balanced display reduces support imbalance in the pre-treatment leads, so it complements rather than replaces the full-window event study when assessing pre-trends.

A separate timing issue is the Families First Coronavirus Response Act (FFCRA), which provided temporary federal paid sick and family leave from April 1 through December 31, 2020. Because FFCRA applied nationally, it narrowed the contrast between state-mandate and non-mandate states during 2020 rather than generating identifying state-level variation. The main specification includes pandemic-period controls for COVID-19 case rates, death rates, and policy responses; in robustness checks, we additionally exclude the 2020 FFCRA window.

Given this structure, we assess the identifying assumption using a set of complementary diagnostics, each aimed at a different threat to interpretation. Event-study estimates examine whether treated and comparison states were already diverging before the coded treatment year and show the timing of the post-treatment response. Leave-one-treated-state-out estimates assess whether the primary care-utilization result is driven by a single adopting state. The pandemic-period exclusion addresses the temporary nationwide FFCRA leave policy. Randomization-inference checks ask whether similarly structured but arbitrary treatment timing within Census divisions would generate estimates as large as the observed one, while subgroup-composition balance checks assess whether mandate timing predicts

observed child or household composition. Finally, TWFE estimates provide a conventional benchmark against the heterogeneity-robust Sun–Abraham estimates, and the YRBSS adds a longer-panel check using adolescent self-reported symptoms rather than parent-reported care or diagnosis. Identification ultimately requires that, absent PSL adoption, treated and comparison states would have followed similar trends in adolescent mental health outcomes, with event-time coefficients interpreted relative to the survey year immediately preceding each state's statutory PSL effective date.

5 Results

5.1 Adolescent Mental Health Outcomes from the NSCH

Figure 1 provides descriptive evidence on raw outcome paths before imposing the regression design. It plots weighted calendar-year means for ever-treated and never-treated NSCH states in the analytic sample. Mental health care use rises over the panel in both groups, with a level shift after 2020, indicating that the outcomes are moving with common calendar-year forces. These raw levels benchmark the magnitude of the estimates below.

(Figure 1 here)

Table 3 presents estimates of the effect of state paid sick leave (PSL) mandates on adolescent mental health care utilization and diagnostic outcomes, estimated using Equation (1). The results indicate that PSL mandates are associated with statistically significant increases in adolescents' engagement with mental health care. The probability that an adolescent received any mental or behavioral health care in the past 12 months increases by 3.0 percentage points ($p < 0.01$), corresponding to a 16.4% increase relative to the untreated baseline mean of 18.2%. Medication use rises by 1.7 percentage points ($p < 0.01$), providing corroborating evidence for increased engagement with clinical care. PSL mandates are also associated with positive diagnosis estimates: the probability of a current depression diagnosis increases by

1.6 percentage points ($p < 0.01$), while the estimated effect on anxiety diagnoses is 1.3 percentage points ($p < 0.05$). We interpret these diagnosis outcomes as complementary evidence because diagnosis rates can rise when previously unidentified need is detected and reported, not only when underlying morbidity worsens.

(Table 3 here)

Figure 2 plots full-window event-study estimates for all four NSCH outcomes, with event time measured relative to statutory PSL adoption. The primary care-utilization outcome provides the clearest dynamic support for the main estimate. The proximate pre-treatment coefficients are small and statistically indistinguishable from zero, the adoption-year coefficient is essentially null, and the estimate jumps to 3.6 percentage points ($p < 0.01$) in the first post-adoption year and remains positive thereafter. This discontinuity between the adoption year and the first post-adoption year is consistent with families needing time to accrue leave, learn about the benefit, and schedule care, and it provides independent support for the one-year-lagged treatment definition used in the static specification rather than a timing convention chosen after the fact. The one visible pre-period deviation for care receipt is a negative lead at $k = -4$, identified off the most thinly supported cohorts and running against rather than toward a positive pre-policy run-up.

The secondary NSCH outcomes move in the same general direction but are dynamically less clean. Medication use follows the care-utilization pattern most closely, with little movement in the adoption year and an increase afterward. Anxiety diagnoses are positive after adoption but estimated less precisely, and the pre-period is not uniformly flat. Depression diagnoses require the most caution, since the pre-period is not flat and a joint Wald test rejects parallel pre-trends in the balanced window ($p < 0.01$). The direction of the violation is nonetheless informative: the depression leads decline steadily as adoption approaches, so treated states were on a relatively falling depression path before PSL. A pre-trend of this sign biases against the positive post-adoption estimate we recover, which makes the depression result conservative rather than mechanically generated by pre-existing divergence. Appendix

Figure A1 restricts the plotted dynamics to cohorts observed for at least three pre-treatment years; in that window, joint Wald tests of the displayed leads do not reject parallel pre-trends for any mental health care ($p = 0.64$) or medication ($p = 0.16$), are marginal for anxiety ($p = 0.07$), and reject for depression ($p < 0.01$). We therefore read the event-study evidence as strongest for completed care receipt, supportive for medication and anxiety, and weakest for depression, and we treat the static depression estimate as corroborating rather than independently identified.

(Figure 2 here)

Table 4 reports exploratory subgroup estimates for the four NSCH mental health outcomes, using the same subgroup columns in each panel. The main message is breadth rather than a sharp demographic gradient. For any mental health care (Panel A), estimates are positive and statistically significant in every subgroup cell except the Black category, which is small and negative, indicating that the average care-utilization effect is not driven by one demographic cell. The medication and diagnosis panels are also predominantly positive, though individually less precise: most cells are positive in each panel, but fewer than half of the depression and anxiety cells are individually significant. Formal equality tests provide some evidence of heterogeneity, but not a single consistent pattern. Race/ethnicity is the dimension that rejects equality most often, doing so for care receipt, anxiety, and depression, while household income rejects for depression and anxiety and parental education rejects for medication. We read the race/ethnicity pattern cautiously. Because this test compares five groups rather than two, it has more dimensions along which coefficients can differ, and rejecting equality indicates only that subgroup coefficients differ, not which group gains more. In each panel the largest estimate falls in the other-or-multiracial category, which is also the one category where the covariate-balance check in Appendix Table A9 finds that mandate timing predicts composition ($p < 0.01$), so part of the measured heterogeneity may reflect compositional differences across treated and comparison states rather than differential policy response. These patterns point to possible heterogeneity, but they do not line up into a single

monotonic gradient by socioeconomic status or family structure. We therefore interpret the subgroup results as descriptive evidence that PSL effects are not concentrated in a single observable group, rather than as evidence of a clear demographic gradient or of changes in disparities in realized care.

(Table 4 here)

The NSCH results show increased engagement with the mental health care system following PSL adoption. Before interpreting this pattern as improved access, however, we must address whether higher diagnosis rates reflect better detection of previously unidentified conditions or a genuine worsening of underlying adolescent mental health. The next subsection uses independent self-reported symptom data from the YRBSS to distinguish between these interpretations.

5.2 Symptom Triangulation Evidence from YRBSS

The YRBSS analysis serves two complementary roles. First, because the NSCH outcomes capture clinical diagnoses and treatment receipt, the estimated increases could reflect either improved detection of previously undiagnosed adolescents or a genuine worsening of underlying mental health. The YRBSS measures self-reported symptoms regardless of clinical contact, so a worsening-mental-health interpretation of the NSCH results would imply a corresponding rise in YRBSS symptoms following PSL adoption. Declining self-reported symptoms alongside higher NSCH care receipt and positive diagnosis estimates is inconsistent with pure worsening, supporting a detection-and-treatment interpretation. Second, the YRBSS panel extends to 2005 and supplies a substantially longer pre-period than the NSCH, using a different instrument, outcome family, and respondent (the adolescent rather than the parent).

Table 5 summarizes Sun–Abraham estimates of PSL effects on three YRBSS measures: self-reported depressive symptoms, suicidal ideation, and suicide planning. Depressive symp-

toms are estimated to decline by 2.2 percentage points ($p < 0.05$) relative to an untreated baseline mean of 31.6 percent. Suicidal ideation declines by 0.8 percentage points and is marginally significant ($p < 0.10$) relative to a 16.9 percent baseline, while suicide planning is negative but imprecisely estimated (-0.6 percentage points). Appendix Figure A2 plots the corresponding event studies as dynamic context. These estimates are noisy, especially for depressive symptoms, where the pre-period is not flat and early post-adoption estimates are mixed before later estimates turn negative. The pre-period leads do, however, drift upward on balance, a direction that works against rather than generates the estimated decline, so the static estimate is conservative relative to the pre-existing trend. We therefore interpret the YRBSS evidence primarily as complementary symptom-side evidence rather than as a standalone dynamic validation exercise. The static YRBSS estimates nonetheless move in the opposite direction of a worsening-mental-health interpretation, especially for depressive symptoms.

(Table 5 here)

5.3 Pandemic Timing and Concurrent Leave Policies

We first assess whether the main NSCH estimates are sensitive to pandemic timing or concurrent leave-policy environments. The 2020 NSCH wave is a natural concern because it overlaps both the acute pandemic disruption to health care access and the temporary federal paid sick leave created by the Families First Coronavirus Response Act (FFCRA). If the main estimate were driven by this unusual policy and health-care environment, excluding the 2020 wave should materially attenuate the result. Instead, Table 6 shows that the primary care-utilization estimate remains positive and statistically significant at 2.7 percentage points ($p < 0.01$). Medication and depression also remain positive and statistically significant, while anxiety remains positive but less precisely estimated.

(Table 6 here)

A second concern is whether the result is confined to PSL states that operate broader paid family and medical leave (PFML) programs. PFML differs from PSL because it typically covers longer absences and may signal a broader paid-leave infrastructure, so it is important to ask whether the estimated PSL effect appears only in states with that additional policy environment. Table 7 splits PSL-treated states by whether active PFML benefits are present at any point in the retained NSCH panel, while retaining never-treated comparison states in each specification. The care-utilization estimate is positive and statistically significant in both policy environments, including 3.1 percentage points among PSL-treated states without active PFML. Medication use is also positive in both samples. The diagnosis estimates are more pronounced in PSL states without active PFML benefits: depression rises by 2.3 percentage points in the without-PFML group, compared with 0.8 percentage points in the with-PFML group, and anxiety follows the same qualitative pattern. In Panel C, the pooled TWFE triple-difference diagnostic is positive and marginally significant for depression when coded as the non-PFML minus PFML contrast (1.5 percentage points; s.e. 0.7 percentage points; $p < 0.10$), while the corresponding differences for the other outcomes are smaller and not statistically significant. These patterns indicate that the PSL estimates are not driven by treated states with active PFML benefits and are consistent with a larger marginal detection role where broader paid-leave infrastructure is absent.

(Table 7 here)

We also examine overlap with broader paid-time-off policies. General PTO mandates are conceptually broader than PSL because they may provide paid time that can be used for illness, family needs, or other purposes, making it harder to attribute changes specifically to paid sick leave. We therefore exclude states with active PTO mandates in the panel (Appendix Table A2). The any-mental-health-care estimate remains close to the baseline estimate at 2.9 percentage points ($p < 0.01$). Medication, depression, and anxiety also remain positive. The result is therefore not driven by states with broad PTO laws.

Across these checks the care-utilization estimate stays between 2.7 and 3.1 percentage points, and it is present in PSL states without active PFML or PTO mandates. The estimated effect is therefore attributable to paid sick leave itself rather than to the pandemic-period policy environment or to co-occurring leave programs.

5.4 Mechanism Evidence: Protected Time and the Conversion of Leave into Care

If PSL raises adolescent mental health care by relaxing parental time constraints, we should see parents reallocate time toward the child's care. Table 8 reports the clearest direct measure: PSL increases the share of families reporting *any* time spent arranging or coordinating the adolescent's health care in the past week—making appointments and locating services—by 3.2 percentage points, a 28% increase relative to the 11.4% baseline ($p < 0.01$). Because the coordination-time measure is categorical and most families report zero time, we decompose the response into weekly care-coordination categories. Most of the increase comes from families reporting less than one hour of weekly coordination time: the estimate for this category is 2.6 percentage points, relative to an 8.0% baseline ($p < 0.01$). Reporting one or more hours of weekly coordination time rises by a smaller 0.6 percentage points, relative to a 3.4% baseline ($p < 0.05$). The category decomposition suggests that the response occurs primarily at the entry margin into low-intensity care coordination rather than through large increases in intensive coordination time. Appendix Figure A3 plots the corresponding event study, estimated on the same employed-caregiver sample, as dynamic description. Three of the four pre-treatment leads are individually indistinguishable from zero.

(Table 8 here)

Table 9 tests whether the coordination-time mechanism concentrates among households more likely to have lacked employer-provided paid sick leave before the mandate, restricting to adolescents with at least one employed caregiver so the comparison stays close to the

population for whom a work-based mandate can operate. We use two proxies for the pre-policy coverage gap: household income and insurance type. Insurance shows the clearest gradient. Among adolescents with public or no insurance, PSL raises any weekly coordination time by 5.1 percentage points from an 11.5 percent baseline, compared with 2.4 percentage points from a 10.6 percent baseline among privately insured adolescents—a response more than twice as large in both absolute and relative terms. The income split runs in the same direction but is more muted: 3.5 percentage points at or below 200 percent of the federal poverty line, against 2.2 percentage points in the top-coded 400 percent group. In both splits the response is concentrated at the under-one-hour margin, consistent with the entry-margin pattern in the pooled estimates. Appendix Table A3 reports formal equality tests of these differences. The unconditional gaps are positive for both proxies. The insurance gap is roughly twice the income gap in magnitude, though neither is estimated precisely enough to reach conventional significance without the no-PFML restriction. Restricting to states without an active Paid Family and Medical Leave program, where PSL is the only source of protected short-duration leave, the insurance gap is statistically significant for both any coordination time and the under-one-hour margin, while the income gap is significant only at the under-one-hour margin, though both income differences are positive and similarly signed. We therefore read both proxies as suggestive, with insurance the more informative of the two, and the overall pattern as consistent with but not decisive for effects concentrating among newly covered families. Both measures remain coarse correlates of individual PSL coverage, and the higher-resource comparison groups may also include caregivers who lacked paid sick leave before the mandate.

(Table 9 here)

This time reallocation operates through short, appointment-day absences rather than broader labor-supply changes. If PSL instead raised care by inducing parents to cut hours or leave employment, the family work-adjustment columns in Table 8 should move. The cut-work-hours estimate is 0.6 percentage points relative to a 4.3% baseline; its 95% confidence

interval allows increases as large as 1.8 percentage points, so moderate changes in hours cannot be fully ruled out. By contrast, the stopped-working estimate is a relatively precise null: -0.4 percentage points relative to a 2.8% baseline, with a 95% confidence interval that rules out increases larger than 0.3 percentage points. Thus, PSL does not appear to increase employment exit. This is the expected signature of short, job-protected leave—roughly 3–7 paid days per year can absorb appointment-day absences without shifting families toward formal reductions in hours or employment exit—and it weighs against a labor-supply or income interpretation of the care gains.

Appendix Table A4 reports broader parent–adolescent engagement outcomes. The engagement index and two problem-solving components are positive, but the estimates are sensitive to an anomalous NY 2019 pre-period dip visible in the raw responses, so we treat them as exploratory. Appendix Table A5 reports related access and caregiving outcomes: unmet mental health need and home health care time are essentially unchanged, preventive visits decline slightly, and reported trouble getting an appointment rises. These checks support interpreting the main mechanism evidence as targeted care coordination rather than a broad shift in family engagement or caregiving intensity.

Taken together, the mechanism evidence describes a coherent chain: PSL frees short protected time, parents reallocate it toward coordinating the adolescent's care, and completed care rises without evidence that families reduce work hours or exit employment to obtain care.

5.5 Additional Validation Checks

Additional diagnostics assess estimator sensitivity, covariate inclusion, influential treated states, random treatment timing, and predetermined covariate balance. We first compare the preferred Sun–Abraham estimates with alternative staggered-adoption benchmarks (Appendix Table A6). The conventional TWFE estimates are positive across the NSCH outcomes and statistically significant for care utilization, medication use, and depression, with

a care-utilization estimate of 2.7 percentage points. The Callaway–Sant'Anna repeated-cross-section estimates are directionally consistent with the preferred estimates across all four NSCH outcomes but are less precise than the Sun–Abraham estimates; care utilization, medication, and depression remain statistically significant, while anxiety is positive but not statistically significant. Sequentially adding control sets yields a similar pattern: the NSCH care-utilization estimate remains between 2.3 and 3.1 percentage points across the demographic, economic, health-supply, COVID, and concurrent-policy specifications, and the depression estimate remains positive and statistically significant (Appendix Table A7). The YRBSS stepwise estimates require state economic controls: with demographics alone all three are indistinguishable from zero, but once economic conditions enter, the depressive-symptom estimate is negative and significant and is nearly identical across the remaining specifications (Appendix Table A8). Because unemployment and income vary substantially across states over 2005–2023 and plausibly correlate with both mandate adoption and youth mental health, we treat the demographic-only column as misspecified.

To examine influence from individual adopters, we drop each treated state in turn (Appendix Table A10). The any-mental-health-care estimate remains positive and statistically significant in every omission, with point estimates ranging from 2.3 to 3.5 percentage points. The medication estimate is similarly stable and statistically significant across all omissions. The diagnosis outcomes are less tightly estimated but remain positive in every omission: depression loses statistical significance only when Arizona is omitted, and anxiety loses statistical significance when Arizona or Washington is omitted. The NSCH results are therefore not driven by the earliest treated state, the newest treated state, or any single treated-state omission.

The randomization-inference exercise asks whether similarly structured but arbitrary adoption timing would generate estimates as large as the observed one (Figure 3). We randomly reassign treatment timing within Census divisions while preserving treated cohort counts and adoption years within each division, then re-estimate the main Sun–Abraham

specification. This design preserves the regional clustering of PSL adoption rather than comparing treated states to arbitrary states nationwide. Across 500 successful placebo assignments, the observed care-utilization estimate of 3.0 percentage points lies above the 95th percentile of the placebo distribution, yielding a right-tail randomization p-value of 0.008.

Covariate-balance checks provide a complementary test of whether mandate timing predicts the nonredundant subgroup-composition indicators used in Table 4 (Appendix Table A9). The expanded checks include age, race/ethnicity, sex, parental education, household income, family structure, and insurance type, omitting complementary binary categories whose coefficients are mechanically redundant. All are statistically indistinguishable from zero except the other-or-multiracial category, with a 0.9 percentage-point estimate ($p < 0.01$), so we report the check transparently and interpret it alongside the randomization-inference, YRBSS symptom, and leave-one-state-out evidence.

6 Discussion and Conclusion

This paper provides, to our knowledge, among the first quasi-experimental evidence that state PSL mandates affect adolescent mental health by increasing engagement with care. Existing quasi-experimental work on PSL and mental health has examined working-age adults (Eisenberg et al., 2025) or prescription fills among Medicaid enrollees (Maclean et al., 2024); neither studies adolescent outcomes or the parental-time channel through which a work-based mandate reaches a dependent child. The primary NSCH estimate shows a 3.0 percentage-point increase in receipt of any mental or behavioral health care, with medication use and parent-reported current diagnoses moving in the same direction. The YRBSS provides an independent symptom-side check: self-reported depressive symptoms decline after PSL adoption, while suicidal ideation and suicide planning also move negative but are less precisely estimated. Taken together, the results point to greater contact with the mental health care system rather than simply more recorded morbidity. This interpretation is reinforced by

validation exercises showing that the care-utilization estimate is stable when excluding the 2020 NSCH wave, which overlaps both the acute COVID-19 disruption to health care access and the temporary federal leave created by the Families First Coronavirus Response Act; excluding states with broad paid-time-off mandates; using alternative staggered-adoption estimators; and dropping treated states one at a time.

A central interpretive question is whether higher NSCH care receipt and positive diagnosis estimates reflect improved detection or a genuine worsening of adolescent mental health. This distinction matters because diagnosed mental, behavioral, and developmental disorders among U.S. children rose substantially over the same period for reasons unrelated to leave policy (Leeb et al., 2024), so a positive diagnosis coefficient is not self-interpreting. The symptom evidence makes a pure worsening interpretation difficult to reconcile with the data: if adolescent mental health were deteriorating after PSL adoption, self-reported symptoms should rise, yet the YRBSS estimates move in the opposite direction. At the same time, the increase in care receipt is proportionally larger than the increase in medication use, consistent with adolescents entering the system through evaluation, counseling, or other non-pharmacological care. Diagnosis outcomes remain complementary rather than primary evidence because they are sensitive to detection and reporting, but the joint pattern of higher care receipt, positive diagnosis estimates, and lower self-reported symptoms is most consistent with improved detection and treatment initiation.

The estimated effect is an intent-to-treat effect for all adolescents in the NSCH sample, not an effect only for children whose parents newly gained PSL coverage. The families most directly affected are a narrower group: working parents covered by a new mandate who previously lacked employer-provided PSL and faced scheduling constraints around adolescent care. Because pre-mandate coverage is not observed in the NSCH, we test this composition logic indirectly, asking whether the coordination-time response is larger where the pre-policy coverage gap was widest (Bureau of Labor Statistics, 2024). Among adolescents with at least one employed caregiver, both proxies point the same way: households with lower income

or with public or no insurance show larger coordination-time responses than their higher-resource counterparts, a difference that is statistically distinguishable for insurance, and more narrowly for income, in states without an active Paid Family and Medical Leave program. We read this as consistent with, but not confirmation of, effects concentrating among newly covered families. We do not convert this estimate into an effect on newly covered families alone. Our estimate averages over all adolescents in the sample, whereas estimates of how much mandates raise paid sick leave coverage are calculated among workers (Callison and Pesko, 2022), and the NSCH does not record which parents previously lacked paid sick leave. The data therefore do not support recovering an effect for that narrower group.

In relative terms, the 16.4% increase in adolescent mental health care receipt is smaller than the 31% increase in Medicaid-funded mental health admissions after ACA Medicaid expansion estimated by Ortega (2023), though it overlaps the lower part of that estimate's confidence interval. This comparison is a scale benchmark rather than a like-for-like effect comparison, since Medicaid expansion operated through insurance coverage whereas PSL operates through parental time. By contrast, policies that reduce financial barriers without directly addressing time constraints have shown limited effects on the extensive margin of mental health treatment, including federal mental health parity mandates (Drake et al., 2019) and minimum wage increases (Kavanagh et al., 2024). These comparisons underscore the central point: parental time is a distinct input into adolescent mental health care access.

The mechanism evidence is consistent with a targeted parental-time account. PSL mandates typically provide only a few paid days per year (Pichler and Ziebarth, 2024), but that amount can matter if the binding margin is the initial evaluation, medication-management visit, or early counseling session that establishes contact with the care system. Consistent with this reading, the coordination response concentrates at the entry margin rather than in sustained caregiving time, and the work-adjustment estimates show no broad labor-supply response—the expected pattern given that U.S. sick-pay mandates have not generated large employment or hours responses (Pichler and Ziebarth, 2017). PSL appears to help some

families cross the threshold into care by making short, appointment-day absences feasible, rather than by inducing sustained reductions in work hours or general changes in family engagement.

Two further splits speak to external validity rather than to the mechanism itself. The positive care estimate appears in PSL-treated states both with and without active PFML benefits, and diagnosis responses are somewhat more pronounced where PFML is absent, consistent with a larger marginal detection role where other paid-leave infrastructure is missing. Demographic subgroup estimates, by contrast, do not map into a single monotonic gradient by income, parental education, race/ethnicity, sex, age, or family structure. Both splits are descriptive: together they indicate that the average effect is not confined to one narrow demographic cell.

Several limitations remain. The NSCH outcomes are parent-reported, and PSL adoption may change which parent responds in multi-parent households; such reporter selection could affect diagnosis reports independent of clinical detection. The YRBSS mitigates this concern by using adolescent self-reports, but it samples enrolled students, so changes in school enrollment or chronic absenteeism could affect the composition of respondents. The NSCH mechanism measures capture reported care coordination and broad work adjustments, not actual leave-taking on appointment days. The NSCH panel also begins in 2016, leaving limited pre-treatment support for some cohorts, even though the event-study diagnostics, leave-one-state-out exercise, randomization inference, and the longer YRBSS panel provide complementary evidence on stability and timing. Finally, even with rich economic, health-supply, COVID, and concurrent-policy controls, residual confounding from other state mental-health investments or progressive policy changes remains possible, and state-level mandate coding masks heterogeneity in local implementation, employer compliance, and enforcement intensity. Future work should examine treatment completion, academic outcomes, and family well-being using administrative claims or school-linked data that can observe care pathways more directly.

The policy implication is therefore not that PSL alone can solve adolescent mental health access gaps. Rather, PSL appears to ease a real time barrier at the point when parents must recognize need, arrange care, and attend visits. Three features of the results bear on how such mandates should be evaluated. First, because the response concentrates at the entry margin into low-intensity coordination rather than in sustained caregiving time, complementary policies should target the friction of initiating care, including scheduling assistance, care navigation, and school-based services that shorten the path from recognizing a problem to being seen (Ali et al., 2019), rather than the capacity to sustain care at home. Second, the mandates studied here provide only a few paid days per year yet still move this margin, while the one-or-more-hour category responds far less strongly; the marginal value of expanding covered days may therefore be smaller for adolescent care initiation than for other uses of leave. Third, we find no evidence that these mandates induce employment exit, which speaks to a common objection in legislative debate over PSL (Pichler and Ziebarth, 2017), although our hours estimate remains too imprecise to rule out moderate adjustments. Although PSL is progressive at the point of legal entitlement, since pre-mandate coverage is concentrated among higher-wage workers (Bureau of Labor Statistics, 2024), our subgroup evidence does not establish that it narrows disparities in realized adolescent care, so the mandate should not be defended on that basis alone. In that sense, PSL is a meaningful but incomplete lever: it can help families make time for care, but it is not a substitute for the mental health system itself.

Taken together, the results identify parental time as a binding and policy-malleable constraint on adolescent mental health care. A few days of protected leave measurably increase whether families initiate care, with symptom evidence that moves in the same direction and no detectable increase in employment exit. Recognizing time as an input to children's health care, alongside insurance coverage and provider supply, broadens the set of levers available for closing adolescent treatment gaps.

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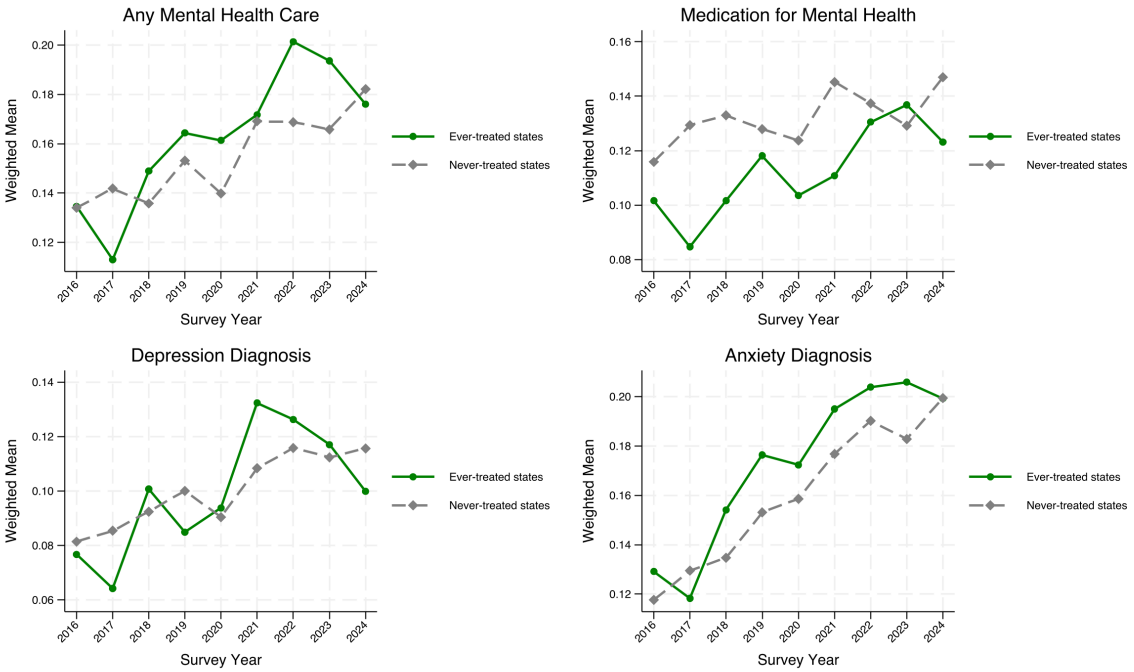
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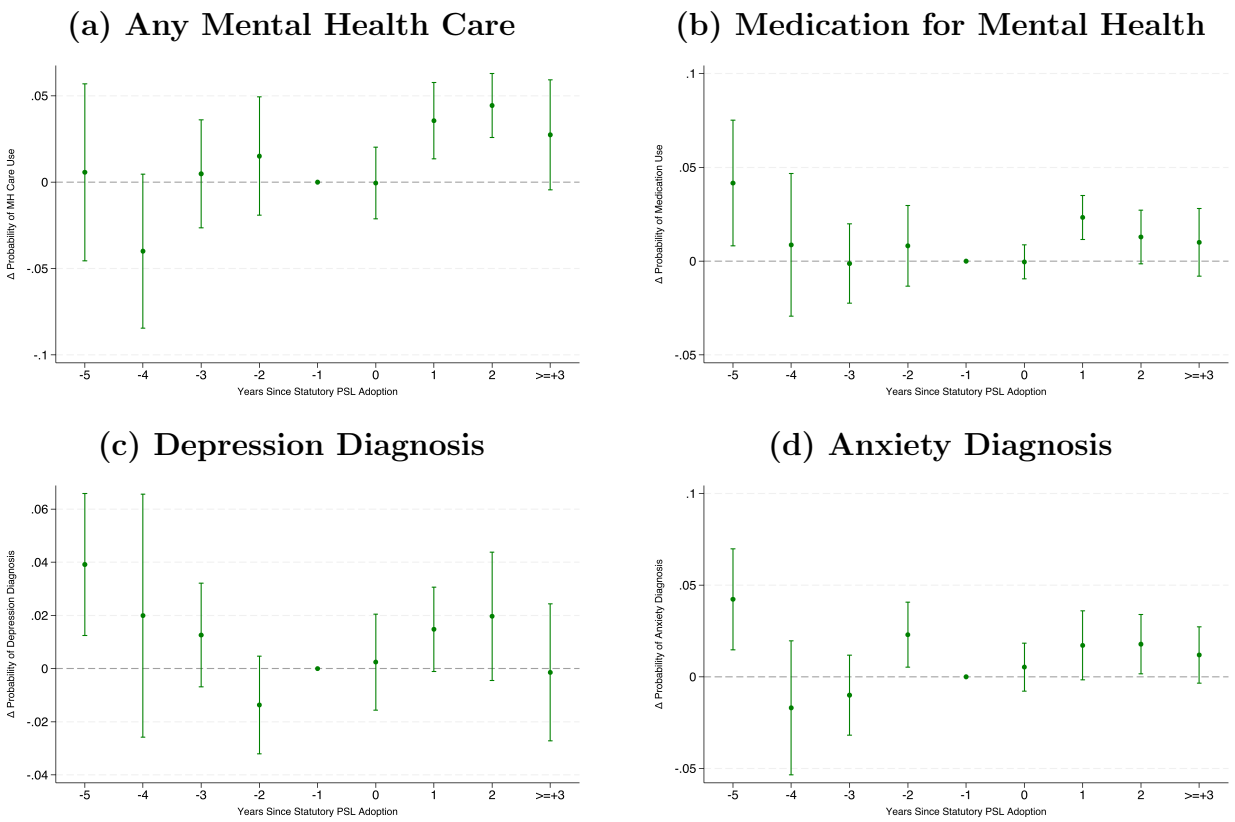
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Figure 1: Descriptive Raw Trends in Adolescent Mental Health Outcomes by State Treatment Status (NSCH, 2016–2024)



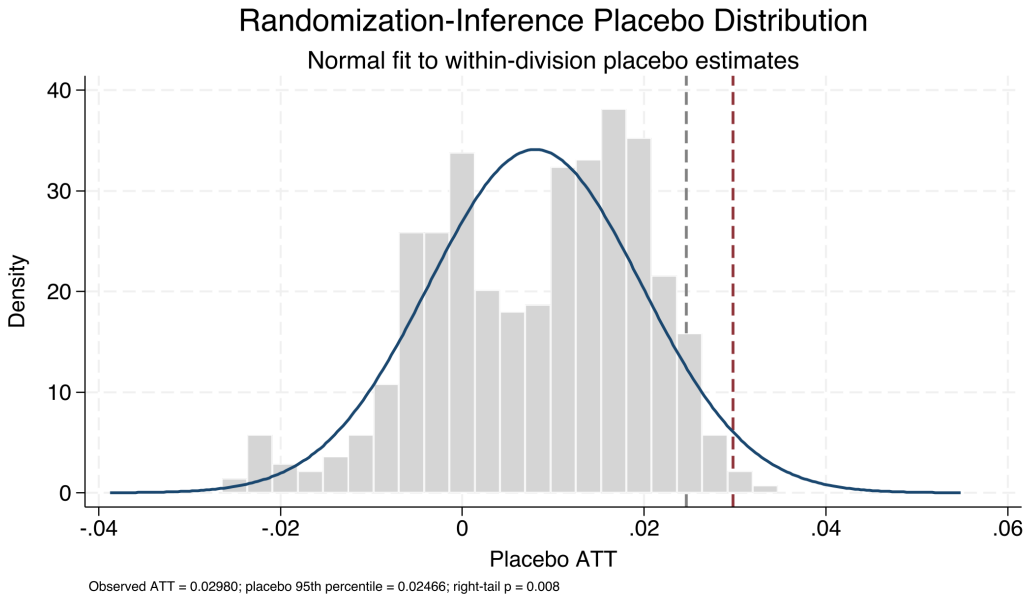
Note: This figure plots weighted calendar-year means for ever-treated NSCH states in the analytic window and never-treated comparison states for four adolescent mental health outcomes: receipt of any mental health care, medication for mental health, parent-reported current depression diagnosis, and parent-reported current anxiety diagnosis. Ever-treated states are retained NSCH states with a statutory PSL adoption year during 2016–2024; never-treated comparison states do not adopt a statewide PSL mandate during the analytic window. These are descriptive raw means; not adjusted for covariates or staggered timing. They are also not composition-adjusted for differences in observed child or household characteristics across state groups. Binary outcomes are reported as proportions; multiply by 100 to convert to percentage points. Means use NSCH sampling weights.

Figure 2: Full-Window Event-Study Estimates for Adolescent Mental Health Outcomes (NSCH, 2016–2024)



Note: This figure presents Sun–Abraham (2021) interaction-weighted event-study estimates for (a) receipt of any mental health care, (b) medication for mental health, (c) parent-reported current depression diagnosis, and (d) parent-reported current anxiety diagnosis. Event time is measured relative to the statutory paid sick leave (PSL) adoption year; $k = -1$ is omitted as the reference period. The full window plots endpoint-binned leads $k \leq -5$ and exact leads $k = -4$ through $k = -2$, the adoption year $k = 0$, post-adoption years $k = +1$ and $k = +2$, and an endpoint-binned lag $k \geq +3$. Estimates use the NSCH sampling weights and the specification described in Table 3. The y-axis reports percentage-point changes in the outcome probability; bars are 95% confidence intervals. The deepest displayed leads are supported mainly by the most recently adopting cohorts and should be interpreted as limited-support diagnostics.

Figure 3: Randomization-Inference Placebo Distribution for Any Mental Health Care (NSCH, 2016–2024)



Note: This figure presents the distribution of placebo ATT estimates from random within-Census-division assignment of PSL treatment timing. The procedure preserves treated cohort counts and adoption years within Census divisions and re-estimates the main NSCH Sun–Abraham static specification for receipt of any mental health care, with state fixed effects, year fixed effects, the main covariate set, NSCH sampling weights, and state-clustered standard errors. The histogram reports the empirical placebo draws, and the curve overlays a normal fit to the placebo mean and standard deviation. The solid vertical line marks the observed ATT; the dashed vertical line marks the 95th percentile of the placebo distribution. Across 500 successful placebo assignments, the observed ATT is 0.0298, the placebo mean is 0.0080, the placebo standard deviation is 0.0117, the placebo 95th percentile is 0.02466, the right-tail randomization p-value is 0.008, and the two-sided absolute-value randomization p-value is 0.008.

Table 1: Descriptive Statistics: NSCH Analytic Sample (2016–2024)

	Full Sample	No Active PSL Mandate	Active PSL Mandate
Outcomes			
Received Any Mental Health Care	0.157	0.154	0.178
Takes Medication for Mental Health	0.128	0.129	0.121
Depression Diagnosis	0.1	0.099	0.111
Anxiety Diagnosis	0.164	0.159	0.193
Demographics			
Female	0.513	0.514	0.511
Child Age (Years)	14.49	14.486	14.512
Number of Adults in Household	2.246	2.232	2.339
Number of Children in Household	2.081	2.081	2.08
Parent Married	0.677	0.675	0.69
Parent Never Married	0.067	0.067	0.064
Parent Divorced	0.113	0.115	0.099
Parent Separated	0.032	0.032	0.034
Parent Unmarried, Partnered	0.049	0.049	0.05
Parent Widowed	0.024	0.025	0.018
Parental Education: Less than High School	0.102	0.102	0.104
Parental Education: High School or GED	0.206	0.209	0.181
Parental Education: Some College or Technical School	0.215	0.218	0.189
Parental Education: College Degree or Higher	0.474	0.466	0.526
Race/Ethnicity: White, non-Hispanic	0.518	0.527	0.458
Race/Ethnicity: Black, non-Hispanic	0.158	0.163	0.123
Race/Ethnicity: Hispanic	0.23	0.221	0.291
Race/Ethnicity: Asian, non-Hispanic	0.038	0.035	0.059
Race/Ethnicity: Other/Multi-racial	0.056	0.055	0.069
Currently Insured	0.916	0.915	0.925
State Controls			
COVID-19 Cumulative Case Rate	0.105	0.096	0.172
COVID-19 Cumulative Death Rate	0.001	0.001	0.002
Medicaid Eligibility: Non-Parents	0.806	0.724	1.38
Medicaid Eligibility: Parents	0.947	0.886	1.38
Community Health Centers per 100k	4.559	4.763	3.13
Primary Care Physicians per 10k	6.006	5.89	6.815
Government COVID Response Index	15.267	14.367	21.57
Housing Price Index	211.389	203.942	263.551
Federal Poverty Level Ratio	251.635	249.385	267.396
Log Per Capita Income	11.046	11.022	11.217
Minimum Wage (Average)	0.463	0.445	0.592
Unemployment Rate	4.471	4.435	4.723
Observations	128,231	110,771	17,460

Note: This table presents weighted descriptive statistics for the NSCH analytic sample. The sample includes adolescents ages 12–17 in the 2016–2024 waves after excluding pre-panel PSL adopter cohorts. Columns distinguish state-years with and without an active state paid sick leave (PSL) mandate. Binary variables are reported as proportions; multiply by 100 to convert to percentages.

Table 2: Treatment Cohorts and Event-Time Support (NSCH and YRBSS)

Cohort (Treatment Year)	States	Pre-Treatment Periods	Post-Treatment Periods	Event-Time Range
Panel A. NSCH (2016–2024, annual waves)				
Pre-panel adopters (excluded)	DC (2009), CT (2012), CA (2015), MA (2015), OR (2016)	0	—	—
2017	Vermont	1	8	$[-1, +7]$
2018	Arizona, Maryland, Washington	2	7	$[-2, +6]$
2019	New Jersey, Rhode Island	3	6	$[-3, +5]$
2021	New York, Colorado	5	4	$[-5, +3]$
2023	New Mexico	7	2	$[-7, +1]$
2024	Minnesota	8	1	$[-8, 0]$
Panel B. YRBSS (2005–2023, biennial waves)				
2009	District of Columbia	2	8	$[-4, +14]$
2012	Connecticut	4	6	$[-7, +11]$
2015	California, Massachusetts	5	5	$[-10, +8]$
2016	Oregon	6	4	$[-11, +7]$
2017	Vermont	6	4	$[-12, +6]$
2018	Arizona, Maryland, Washington	7	3	$[-13, +5]$
2019	New Jersey, Rhode Island	7	3	$[-14, +4]$
2021	New York, Colorado	8	2	$[-16, +2]$
2023	New Mexico	9	1	$[-18, +0]$

Note: This table reports the treated cohorts contributing to each survey's estimation sample and the event-time support available in each panel. For the NSCH (Panel A), the year shown is the statutory PSL adoption year under the post-July coding rule, and event-time ranges are expressed relative to that year. The static NSCH ATT applies a one-year lag to the treatment indicator, so each state's coded treatment year in the static specification is one year later than the statutory year shown. Panel A covers the annual NSCH waves from 2016–2024; the five pre-panel adopters (DC, CT, CA, MA, OR) have no NSCH pre-treatment observations and are excluded from the preferred NSCH Sun–Abraham sample. Minnesota's statutory adoption falls in the final NSCH wave (2024); under the one-year lag, its coded treatment year (2025) falls outside the panel, so Minnesota contributes pre-treatment support but no post-treatment observations to the static ATT. For the YRBSS (Panel B), treatment year is the first biennial survey period in which the mandate is in effect under the same post-July coding rule. Panel B covers the 2005–2023 YRBSS waves; early adopters excluded from the NSCH analysis contribute meaningful pre-treatment support in YRBSS.

Table 3: Effects of Paid Sick Leave Mandates on Adolescent Mental Health Outcomes (NSCH, 2016–2024)

	(1) Any Mental Health Care	(2) Medication for Mental Health Issues	(3) Depression Diagnosis	(4) Anxiety Diagnosis
PSL Mandate	0.0298***	0.0171***	0.0157***	0.0125**
	(0.00686)	(0.00541)	(0.00573)	(0.00543)
Baseline Mean	0.182	0.155	0.117	0.194
Observations	123,372	122,210	123,418	123,351
R-squared	0.029	0.018	0.034	0.040

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates of state paid sick leave (PSL) mandates on adolescents' mental health care utilization and parent-reported current diagnoses. The treatment indicator is lagged one year relative to statutory adoption. Each column reports a separate regression with state and year fixed effects. Covariates include adolescent age, sex, race/ethnicity, parental education and marital status, number of adults and children in the household, state minimum wage, unemployment rate, Medicaid eligibility thresholds, health care provider supply, the Housing Price Index, log per capita income, COVID-19 case and death rates, the Oxford Government Response Index, active paid family and medical leave (PFML), the post-2014 school-based Medicaid expansion, and active PTO mandates. Household income and insurance type are excluded because they may respond to PSL adoption. Estimates use NSCH sampling weights; standard errors clustered at the state level are reported in parentheses. Baseline mean is the outcome mean among untreated observations. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 4: Effects of Paid Sick Leave Mandates on Adolescent Mental Health Outcomes by Subgroup (NSCH, 2016–2024)

	Age		Race/Ethnicity					Sex		Parental Education		Household Income		Family Structure	
	(1) 12–14	(2) 15–17	(3) Asian	(4) Black	(5) Hispanic	(6) Other/Multi	(7) White	(8) Male	(9) Female	(10) No College	(11) College+	(12) ≤200% FPL	(13) >200% FPL	(14) Single	(15) Two-Parent
Panel A: Received Any Mental Health Care															
PSL Mandate	0.0364***	0.0238	0.0525**	-0.0142	0.0441***	0.0751***	0.0233**	0.0378***	0.0206***	0.0429**	0.0147**	0.0246**	0.0310***	0.0262**	0.0295***
	(0.00969)	(0.0160)	(0.0221)	(0.0171)	(0.0124)	(0.0255)	(0.0102)	(0.0119)	(0.00516)	(0.0160)	(0.00606)	(0.00989)	(0.00717)	(0.0130)	(0.00855)
Observations	54,958	68,414	6,150	8,275	14,653	8,195	86,099	59,749	63,623	49,445	73,927	33,476	89,896	30,613	92,759
R-squared	0.029	0.032	0.037	0.033	0.037	0.053	0.028	0.042	0.020	0.028	0.033	0.036	0.030	0.033	0.024
Equality-test p-value	0.584				0.009			0.165		0.160		0.508		0.839	
Panel B: Medication for Mental Health															
PSL Mandate	0.0253**	0.00972	0.0182	0.0106	0.00439	0.0537**	0.0159**	0.0185*	0.0159*	0.0260***	0.00766	0.0147*	0.0177**	0.0311***	0.00965
	(0.00975)	(0.0114)	(0.0158)	(0.0203)	(0.0123)	(0.0210)	(0.00739)	(0.0107)	(0.00796)	(0.00633)	(0.00763)	(0.00852)	(0.00703)	(0.00990)	(0.00695)
Observations	54,489	67,721	6,063	8,204	14,460	8,120	85,363	59,233	62,977	48,885	73,325	33,104	89,106	30,325	91,885
R-squared	0.021	0.025	0.040	0.032	0.024	0.028	0.012	0.029	0.019	0.022	0.018	0.030	0.016	0.019	0.017
Equality-test p-value	0.403				0.272			0.863		0.049		0.799		0.104	
Panel C: Depression Diagnosis															
PSL Mandate	0.0130**	0.0187*	0.0243*	-0.0107	0.0219	0.0633***	0.0102	0.0162	0.0147	0.0131	0.0179***	-0.00257	0.0298***	0.0164	0.0145**
	(0.00584)	(0.0110)	(0.0124)	(0.0237)	(0.0146)	(0.0211)	(0.00760)	(0.0138)	(0.00887)	(0.0124)	(0.00663)	(0.0122)	(0.00673)	(0.0195)	(0.00656)
Observations	54,996	68,422	6,149	8,279	14,680	8,187	86,123	59,766	63,652	49,462	73,956	33,511	89,907	30,613	92,805
R-squared	0.022	0.033	0.043	0.032	0.042	0.065	0.036	0.044	0.023	0.036	0.034	0.046	0.031	0.034	0.027
Equality-test p-value	0.680				0.044			0.941		0.756		0.034		0.935	
Panel D: Anxiety Diagnosis															
PSL Mandate	0.00798	0.0170*	0.0277**	-0.0246	0.0212	0.0676***	0.00165	0.0264**	-0.00169	0.00238	0.0193**	-0.0159	0.0282***	0.0273**	0.00745
	(0.00588)	(0.00933)	(0.0119)	(0.0206)	(0.0143)	(0.0196)	(0.00862)	(0.0131)	(0.0130)	(0.00738)	(0.00912)	(0.0106)	(0.00851)	(0.0124)	(0.00889)
Observations	54,948	68,403	6,155	8,267	14,662	8,186	86,081	59,729	63,622	49,433	73,918	33,493	89,858	30,613	92,738
R-squared	0.030	0.049	0.053	0.031	0.038	0.056	0.033	0.049	0.025	0.044	0.040	0.055	0.037	0.056	0.035
Equality-test p-value	0.431				0.003			0.249		0.188		0.008		0.292	

Note: This table presents exploratory Sun–Abraham (2021) interaction-weighted ATT estimates for adolescent mental health outcomes by subgroup. Each estimate is from a separate regression on the subgroup indicated by the column header. Equality-test p-values are from auxiliary pooled Sun–Abraham specifications with subgroup-specific PSL post indicators and subgroup-specific state fixed effects, year fixed effects, and control slopes; p-values test equality within each grouped column block. Estimates are suppressed for subgroup cells with fewer than 50 observations. Specification, controls, fixed effects, weights, and clustering follow Table 3. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Effects of Paid Sick Leave Mandates on Youth Mental Health Outcomes (YRBSS, 2005–2023)

	(1) Depression	(2) Suicidal Ideation	(3) Suicide Planning
PSL Mandate	-0.0216**	-0.00766*	-0.00585
	(0.00798)	(0.00411)	(0.00546)
Baseline Mean	0.316	0.169	0.139
Observations	986283	886329	960186
R-squared	0.054	0.022	0.016

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates of state PSL mandates on youth self-reported mental health outcomes. The treatment indicator is lagged relative to statutory adoption. Each column reports a separate regression with state and year fixed effects. Covariates include age, sex, race/ethnicity, state minimum wage, unemployment rate, Medicaid eligibility thresholds, health care provider supply, the Housing Price Index, log per capita income, COVID-19 case and death rates, and the Oxford Government Response Index. Estimates use YRBSS sampling weights; standard errors clustered at the state level are reported in parentheses. Baseline mean is the outcome mean among untreated observations. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Pandemic-Period Robustness: Effects of Paid Sick Leave Mandates on Adolescent Mental Health Outcomes (NSCH, 2016–2024)

	(1) Any Mental Health Care	(2) Medication for Mental Health	(3) Depression Diagnosis	(4) Anxiety Diagnosis
PSL Mandate	0.0273***	0.0165**	0.0133**	0.0100*
	(0.0078)	(0.0063)	(0.0061)	(0.0052)
Baseline Mean	0.155	0.129	0.100	0.160
Observations	108,746	107,717	108,787	108,735
R-squared	0.030	0.018	0.034	0.042

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates for the main NSCH specification. Estimates exclude the 2020 NSCH wave, which overlaps with the FFCRA temporary federal paid-leave window. All other specification elements—covariates, fixed effects, weights, and clustering—follow Table 3. Baseline mean is the outcome mean among untreated observations in the restricted sample. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: Effects of Paid Sick Leave Mandates by PFML Policy Environment (NSCH, 2016–2024)

	(1) Any Mental Health Care	(2) Medication for Mental Health	(3) Depression Diagnosis	(4) Anxiety Diagnosis
Panel A: PSL States without PFML				
PSL Mandate	0.0310***	0.0111*	0.0229***	0.0136**
	(0.0071)	(0.0057)	(0.0056)	(0.0062)
Baseline Mean	0.183	0.158	0.119	0.195
Observations	108,485	107,490	108,525	108,464
R-squared	0.028	0.017	0.034	0.038
Panel B: PSL States with PFML				
PSL Mandate	0.0238***	0.0187***	0.0076	0.0093
	(0.0052)	(0.0058)	(0.0050)	(0.0070)
Baseline Mean	0.182	0.157	0.117	0.195
Observations	113,461	112,382	113,504	113,440
R-squared	0.029	0.018	0.034	0.040
Panel C: Pooled TWFE Difference (without PFML minus with PFML)				
PSL × Without-PFML Environment	0.0061	-0.0079	0.0146*	0.0047
	(0.0137)	(0.0143)	(0.0074)	(0.0120)
Baseline Mean	0.182	0.155	0.117	0.194
Observations	123,372	122,210	123,418	123,351
R-squared	0.029	0.018	0.034	0.040

Note: This table presents policy-environment sensitivity checks for PSL-treated states with and without active state paid family and medical leave (PFML) benefits during the retained NSCH panel. Panels A and B present Sun–Abraham (2021) interaction-weighted ATT estimates from separate split-sample regressions. The "PSL states without PFML" sample retains Arizona, Maryland, New Mexico, and Vermont as treated states plus all never-treated comparison states. The "PSL states with PFML" sample retains Colorado, New Jersey, New York, Rhode Island, and Washington as treated states plus all never-treated comparison states. Minnesota contributes no post-treatment observations under the one-year-lagged treatment coding and remains in the comparison pool. Panel C presents the coefficient on the interaction between PSL treatment and an indicator for the non-PFML policy environment from a pooled TWFE triple-difference regression; positive coefficients indicate that the PSL estimate is larger in non-PFML states than in PFML states. This exercise assesses whether the PSL estimates are driven by treated states with active PFML benefits and provides descriptive evidence on differences across paid-leave policy environments. All specification elements follow Table 3. Baseline mean is the outcome mean among untreated observations. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 8: Effects of Paid Sick Leave Mandates on Care Coordination and Parental Work Adjustment (NSCH, 2016–2024)

	(1) Any Weekly Care Coordination Time	(2) Under 1 Hour Weekly Care Coordination	(3) 1 or More Hours Weekly Care Coordination	(4) Cut Work Hours	(5) Stopped Working
PSL Mandate	0.0320***	0.0262***	0.00583**	0.00607	-0.00391
	(0.00551)	(0.00482)	(0.00280)	(0.00602)	(0.00335)
Baseline Mean	0.114	0.080	0.034	0.043	0.028
Observations	122,075	122,075	122,075	122,473	122,721
R-squared	0.009	0.009	0.004	0.007	0.008

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates of state PSL mandates on parent-reported care-coordination and work-adjustment outcomes. Column (1) is an indicator for any time spent arranging or coordinating the child's health care over the past week. The source coordination-time measure is categorical; columns (2)–(3) are mutually exclusive indicators for reporting less than one hour and one or more hours of weekly care-coordination time, respectively. Columns (4)–(5) are family work-adjustment outcomes included as alternative-interpretation checks. Related engagement, access, and caregiving outcomes are reported in the appendix. Specification, controls, fixed effects, weights, and clustering follow Table 3. Baseline mean is the outcome mean among untreated observations. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

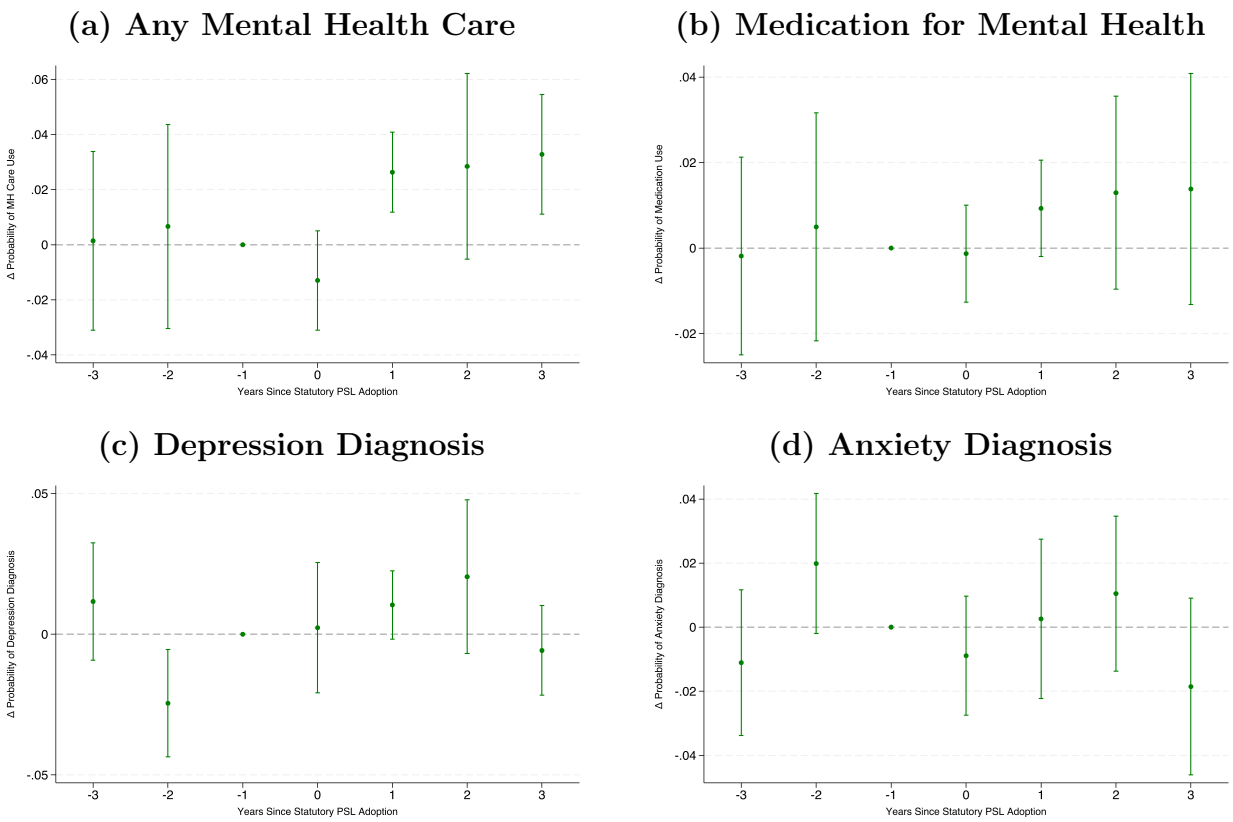
Table 9: Effects of Paid Sick Leave Mandates on Weekly Care Coordination Among Employed-Caregiver Households (NSCH, 2016–2024)

	Any Weekly Care Coordination Time	Under 1 Hour Weekly Care Coordination	1 or More Hours Weekly Care Coordination
Panel A. Household income among employed-caregiver households			
Sample: income at or below 200% FPL			
PSL Mandate	0.0351***	0.0240**	0.0111
	(0.0122)	(0.0092)	(0.0093)
Baseline mean	0.105	0.072	0.034
Observations	26,819	26,819	26,819
Sample: income top-coded at 400% FPL			
PSL Mandate	0.0217***	0.0145**	0.0073
	(0.0059)	(0.0057)	(0.0058)
Baseline mean	0.117	0.090	0.027
Observations	50,969	50,969	50,969
Panel B. Insurance type among employed-caregiver households			
Sample: public insurance or uninsured			
PSL Mandate	0.0512***	0.0411***	0.0101**
	(0.0165)	(0.0148)	(0.0049)
Baseline mean	0.115	0.078	0.037
Observations	22,555	22,555	22,555
Sample: private insurance			
PSL Mandate	0.0236***	0.0198***	0.0038
	(0.0047)	(0.0043)	(0.0028)
Baseline mean	0.106	0.081	0.025
Observations	89,004	89,004	89,004

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates for care-coordination outcomes by household income and insurance type among adolescents with at least one employed caregiver. The employed-caregiver restriction is a coarse policy-relevance restriction, not a measure of newly gained PSL coverage or actual leave use. The outcome columns measure any reported care-coordination time in the past week, less than one hour of weekly care-coordination time, and one or more hours of weekly care-coordination time. The income split compares households at or below 200% of the federal poverty line with the top-coded 400% FPL group; the public-use NSCH income-to-poverty measure is capped at 400%, so this group should not be interpreted as households strictly above 400% FPL. The insurance split compares adolescents with public insurance or no insurance to those with private insurance. Because insurance type is measured contemporaneously we treat it as descriptive, though mandate timing does not predict insurance composition in the balance checks. Baseline mean is the weighted outcome mean among untreated observations in the corresponding sample. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

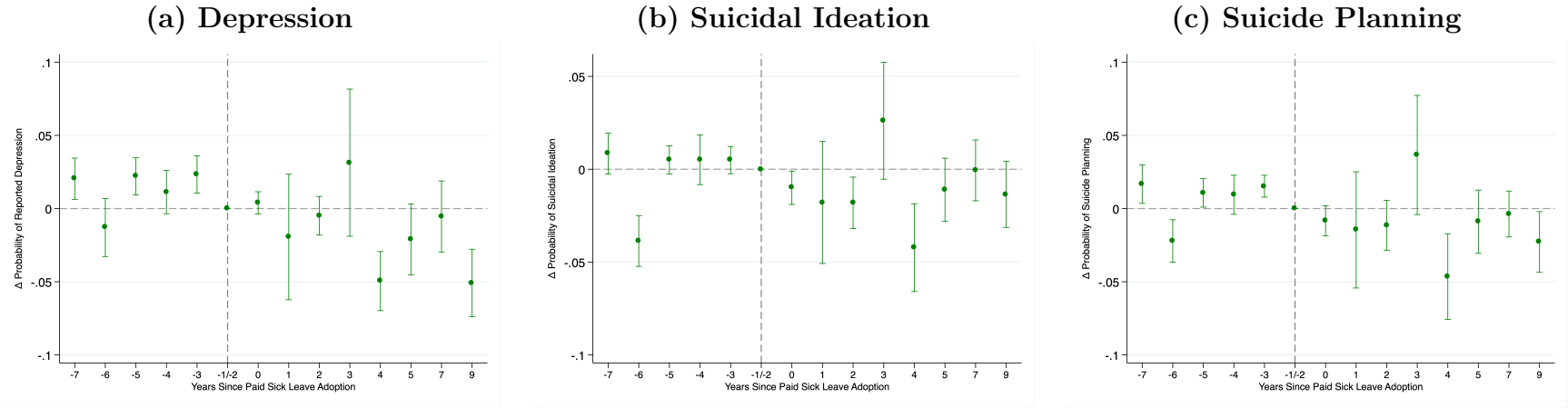
Appendix Figures and Tables

Figure A1: Balanced-Window Event-Study Estimates for Adolescent Mental Health Outcomes (NSCH, 2016–2024)



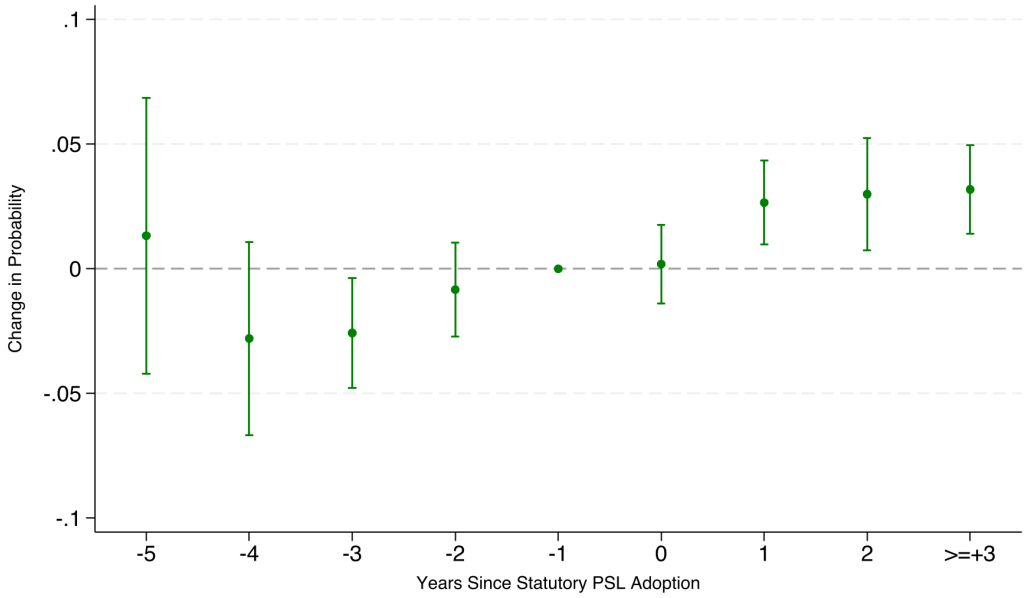
Note: This figure presents Sun–Abraham (2021) interaction-weighted event-study estimates for (a) receipt of any mental health care, (b) medication for mental health, (c) parent-reported current depression diagnosis, and (d) parent-reported current anxiety diagnosis. Event time is measured relative to the statutory paid sick leave (PSL) adoption year; $k = -1$ is omitted as the reference period. The balanced window retains cohorts observed for at least three pre-treatment years and plots leads $k = -3$ and $k = -2$, the adoption year $k = 0$, post-adoption years $k = +1$ and $k = +2$, and an endpoint-binned lag $k \geq +3$. Estimates use the NSCH sampling weights and the specification described in Table 3. The y-axis reports percentage-point changes in the outcome probability; bars are 95% confidence intervals. Joint Wald χ^2 tests of the displayed pre-treatment leads do not reject parallel pre-trends for any mental health care ($p = 0.64$) or medication ($p = 0.16$), are marginal for anxiety ($p = 0.07$), and reject for depression ($p < 0.01$); we therefore treat depression as complementary evidence.

Figure A2: Event-Study Estimates for Youth Mental Health Outcomes (YRBSS, 2005–2023)



Note: This figure presents Sun–Abraham (2021) interaction-weighted event-study estimates for youth self-reported mental health outcomes: (a) depressive symptoms, (b) suicidal ideation, and (c) suicide planning. Event time is measured relative to PSL adoption; $k = -1$ is omitted as the reference period. Because YRBSS is biennial, each event-time unit corresponds to one two-year survey cycle, and the reference period is the last pre-treatment survey wave rather than a literal one-calendar-year lead. Estimates use the YRBSS sampling weights and the specification described in Table 5. The y-axis reports percentage-point changes in the outcome probability; bars are 95% confidence intervals.

Figure A3: Full-Window Event-Study Estimates for Any Care-Coordination Time (NSCH, 2016–2024)



Note: This figure presents Sun–Abraham (2021) interaction-weighted event-study estimates for the probability that a family reports any time spent arranging or coordinating the adolescent's health care, restricted to adolescents with at least one employed caregiver. Event time is measured relative to statutory PSL adoption; $k = -1$ is omitted as the reference period. The figure shows exact event time $k = -5$ through $k = +2$, drops event times $k \leq -6$, and bins event times $k \geq +3$. Estimates use the NSCH sampling weights and the specification described in Table 8. The y-axis reports percentage-point changes in the outcome probability; bars are 95% confidence intervals. The plot is included as descriptive dynamic context rather than as the primary mechanism test.

Table A1: Descriptive Statistics: YRBSS Analytic Sample (2005–2023)

	Full Sample	**No Active PSL Mandate**	**Active PSL Mandate**
Outcomes			
Self-Reported Depression	0.308	0.307	0.312
Suicidal Ideation	0.168	0.167	0.169
Suicide Planning	0.137	0.137	0.139
Demographics			
Female	0.488	0.488	0.489
Age 16 or Older	0.626	0.626	0.624
Race/Ethnicity: White, non-Hispanic	0.546	0.511	0.578
Race/Ethnicity: Black, non-Hispanic	0.164	0.157	0.182
Race/Ethnicity: Hispanic	0.210	0.188	0.260
Race/Ethnicity: Asian, non-Hispanic	0.031	0.029	0.035
Race/Ethnicity: Other/Multi-racial	0.048	0.048	0.050
Observations (depression regression)	1,064,699	—	—

Note: This table presents weighted descriptive statistics for the YRBSS analytic sample. The sample includes high-school students in participating state YRBSS waves from 2005–2023. Columns distinguish state-years with and without an active state PSL mandate. Binary variables are reported as proportions; multiply by 100 to convert to percentages. The full-sample observation count corresponds to the main depressive-symptoms regression; sample sizes vary across outcomes because of item nonresponse.

Table A2: PTO-State Exclusion Robustness: Effects of Paid Sick Leave Mandates on Adolescent Mental Health Outcomes (NSCH, 2016–2024)

	(1) Any Mental Health Care	(2) Medication for Mental Health	(3) Depression Diagnosis	(4) Anxiety Diagnosis
PSL Mandate	0.0285***	0.0137**	0.0136*	0.0122**
	(0.00731)	(0.00509)	(0.00674)	(0.00582)
Baseline Mean	0.181	0.156	0.117	0.192
Observations	113,164	112,102	113,200	113,151
R-squared	0.028	0.018	0.034	0.040

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates after excluding states that ever have an active general paid-time-off mandate in the sample period: Illinois, Maine, Michigan, and Nevada. Because these states are excluded, the active PTO mandate control is omitted from this robustness specification; all other specification elements follow Table 3. Baseline mean is the outcome mean among untreated observations. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3: Coordination-Time Response by Policy-Bite Proxy: Subgroup Differences Among Employed-Caregiver Households (NSCH, 2016–2024)

	Any Weekly Care Coordination Time	Under 1 Hour Weekly Care Coordination	1 or More Hours Weekly Care Coordination
Income (≤200% vs. 400% FPL)	0.0133	0.00950	0.00384
	(0.0125)	(0.0100)	(0.0120)
N (low-resource)	26,819	26,819	26,819
N (high-resource)	50,969	50,969	50,969
Income, no-PFML states only	0.0177	0.0227**	-0.00506
	(0.0107)	(0.01000)	(0.0121)
N (low-resource)	24,749	24,749	24,749
N (high-resource)	46,225	46,225	46,225
Insurance (public/uninsured vs. private)	0.0276	0.0213	0.00628
	(0.0173)	(0.0153)	(0.00564)
N (low-resource)	22,555	22,555	22,555
N (high-resource)	89,004	89,004	89,004
Insurance, no-PFML states only	0.0390***	0.0337***	0.00528
	(0.0104)	(0.00712)	(0.00868)
N (low-resource)	20,788	20,788	20,788
N (high-resource)	81,744	81,744	81,744

Note: Each cell reports the estimated difference in Sun–Abraham (2021) interaction-weighted ATT coefficients between the two subgroups named in the row label (lower-resource group minus higher-resource group), not the level of either group's own coefficient, using pooled specifications with subgroup-interacted state and year fixed effects and covariates among adolescents with at least one employed caregiver. Income rows compare households at or below 200% of the federal poverty line to the top-coded 400% FPL group; the public-use NSCH income-to-poverty measure is capped at 400%, so this group should not be interpreted as households strictly above 400% FPL. Insurance rows compare adolescents with public insurance or no insurance to those with private insurance and are descriptive because insurance type is measured contemporaneously, though mandate timing does not predict insurance composition in the balance checks. "No-PFML" rows restrict the sample to states without an active Paid Family and Medical Leave program at the time of the survey. N (low-resource) and N (high-resource) report the estimation sample sizes for the two subgroups being compared. Standard errors are in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A4: Effects of Paid Sick Leave Mandates on Parent–Adolescent Engagement (NSCH, 2016–2024)

	(1) Engagement Index (SD)	(2) Family Meals Together	(3) Shares Ideas / Talks About Things	(4) Talks About Concerns	(5) Works Together to Solve Problems
PSL Mandate	0.0295*	0.0626	-0.00635	0.0230*	0.0149*
	(0.0159)	(0.0417)	(0.00607)	(0.0122)	(0.00789)
Baseline Mean	-0.003	4.460	0.940	0.865	0.873
Observations	123,654	123,168	122,591	122,898	122,690
R-squared	0.029	0.049	0.007	0.023	0.022

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates for parent–adolescent engagement outcomes. Column (1) reports a standardized index that averages the four components after standardizing each item to the never-treated control distribution following Anderson (2008); columns (2)–(5) report the components separately. These estimates are interpreted as exploratory because the index and problem-solving components are sensitive to an anomalous NY 2019 pre-period dip visible in the raw responses. Specification, controls, fixed effects, weights, and clustering follow Table 3. Baseline mean is the outcome mean among untreated observations. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A5: Effects of Paid Sick Leave Mandates on Access and Caregiving Outcomes (NSCH, 2016–2024)

	(1) Unmet Mental Health Need	(2) Home Health Care Time	(3) Trouble Getting an Appointment	(4) Facility Not Open When Needed	(5) Preventive Check-Up Visit
PSL Mandate	0.00295	0.00206	0.0120***	0.00189	-0.00685*
	(0.00461)	(0.0253)	(0.00289)	(0.00151)	(0.00394)
Baseline Mean	0.031	0.304	0.021	0.008	0.943
Observations	123,372	121,408	123,452	123,437	101,292
R-squared	0.006	0.006	0.008	0.006	0.014

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates for access and caregiving outcomes. Unmet mental health need, trouble getting an appointment, facility not open when needed, and preventive check-up visit are parent-reported binary indicators. Home health care time is a weekly-hours midpoint scale constructed from the NSCH question on providing health care at home. Specification, controls, fixed effects, weights, and clustering follow Table 3. Baseline mean is the outcome mean among untreated observations. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A6: Alternative Estimator Benchmarks: Effects of Paid Sick Leave Mandates on Adolescent Mental Health Outcomes (NSCH, 2016–2024)

	(1) Any Mental Health Care	(2) Medication for Mental Health Issues	(3) Depression Diagnosis	(4) Anxiety Diagnosis
Panel A: Conventional TWFE				
PSL Mandate	0.0290***	0.0156**	0.0135**	0.00907
	(0.00669)	(0.00688)	(0.00605)	(0.00704)
Baseline Mean	0.154	0.128	0.099	0.160
Observations	123,372	122,210	123,418	123,351
R-squared	0.029	0.018	0.034	0.040
Panel B: Callaway–Sant'Anna				
PSL Mandate	0.0306**	0.0182*	0.0174*	0.00940
	(0.0127)	(0.0107)	(0.0101)	(0.0138)
Baseline Mean	0.154	0.128	0.099	0.160
Observations	127,558	126,306	127,700	127,623

Note: This table presents alternative estimator benchmarks for the NSCH outcomes. Panel A reports conventional two-way fixed effects (TWFE) estimates with state and year fixed effects and the full Table 3 covariate set. Panel B reports Callaway–Sant'Anna (2021) doubly robust repeated-cross-section ATT estimates using not-yet-treated observations as controls and aggregating group-time effects with the simple ATT aggregation. The treatment indicator is lagged one year relative to statutory adoption. Both panels use NSCH sampling weights and state-clustered standard errors. Baseline mean is the outcome mean among untreated observations. R-squared is reported only for Panel A. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A7: Stepwise Robustness: Effects of Paid Sick Leave Mandates on Adolescent Mental Health Outcomes (NSCH, 2016–2024)

	(1)	(2)	(3)	(4)
Panel A: Received Any Mental Health Care				
PSL Mandate	0.0231***	0.0304***	0.0312***	0.0285***
	(0.00392)	(0.00514)	(0.00511)	(0.00686)
State FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Demographics/HH	YES	YES	YES	YES
State Economic		YES	YES	YES
Health Supply			YES	YES
COVID Controls				YES
Policy/PTO Controls				YES
Observations	123,372	123,372	123,372	123,372
Panel B: Depression Diagnosis				
PSL Mandate	0.0118**	0.0153***	0.0156***	0.0150**
	(0.00447)	(0.00446)	(0.00475)	(0.00559)
State FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Demographics/HH	YES	YES	YES	YES
State Economic		YES	YES	YES
Health Supply			YES	YES
COVID Controls				YES
Policy/PTO Controls				YES
Observations	123,418	123,418	123,418	123,418

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates with sequentially added controls. Panel A examines receipt of any mental health care, and Panel B examines parent-reported current depression diagnosis. All columns include state and year fixed effects. Column (1) includes demographic and household controls; column (2) adds state economic conditions; column (3) adds health care supply measures; column (4) adds COVID-related controls and concurrent-policy/PTO controls. Estimates use NSCH sampling weights; standard errors clustered at the state level are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A8: Stepwise Robustness: Effects of Paid Sick Leave Mandates on Youth Mental Health Outcomes (YRBSS, 2005–2023)

	(1)	(2)	(3)	(4)
Panel A: Self-Reported Depression				
PSL Mandate	-0.00475	-0.0203***	-0.0208***	-0.0202**
	(0.00679)	(0.00634)	(0.00660)	(0.00801)
Observations	1097458	986283	986283	986283
Panel B: Suicidal Ideation				
PSL Mandate	0.00407	-0.00554	-0.00524	-0.00714*
	(0.00319)	(0.00451)	(0.00401)	(0.00419)
Observations	981388	886329	886329	886329
Panel C: Suicide Planning				
PSL Mandate	0.00174	-0.00422	-0.00438	-0.00572
	(0.00400)	(0.00521)	(0.00538)	(0.00578)
State FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
Demographics/HH	YES	YES	YES	YES
State Economic		YES	YES	YES
Health Supply			YES	YES
COVID Controls				YES
Observations	1063677	960186	960186	960186

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates with sequentially added controls. Panel A examines depressive symptoms, Panel B examines suicidal ideation, and Panel C examines suicide planning. All columns include state and year fixed effects. Column (1) includes demographic controls; column (2) adds state economic conditions; column (3) adds health care supply measures; column (4) adds COVID-related controls. Estimates use YRBSS sampling weights; standard errors clustered at the state level are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A9: Covariate-Balance and Subgroup-Composition Checks (NSCH, 2016–2024)

Outcome	ATT	SE	p-value
Age 12–14	-0.00127	0.00870	0.885
Asian, non-Hispanic	-0.00103	0.00611	0.867
Black, non-Hispanic	0.00256	0.00986	0.796
Hispanic	-0.00108	0.00625	0.864
Other/Multi	0.00851	0.00271	0.003
White, non-Hispanic	-0.00897	0.00812	0.276
Male	-0.00686	0.01126	0.545
No college	0.00182	0.00967	0.852
Income 0–200% FPL	-0.00308	0.00650	0.638
Single-parent family	0.00500	0.00906	0.584
Public or no insurance	0.01267	0.00965	0.196

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates using nonredundant subgroup-composition indicators as outcomes. The checks assess whether PSL mandate timing predicts observed age, race/ethnicity, sex, parental-education, household-income, family-structure, or insurance composition. Complementary binary categories are omitted when their coefficients are mechanically redundant. Household income and insurance type are measured contemporaneously and are therefore interpreted as subgroup-composition diagnostics rather than predetermined controls. All models use the main NSCH specification with state fixed effects, year fixed effects, the relevant covariate set, NSCH sampling weights, and state-clustered standard errors.

Table A10: Leave-One-State-Out Robustness: Effects of Paid Sick Leave Mandates on Adolescent Mental Health Outcomes (NSCH, 2016–2024)

State Dropped	Any Mental Health Care	Medication for Mental Health	Depression Diagnosis	Anxiety Diagnosis
None (Baseline)	0.0298***	0.0171***	0.0157***	0.0125**
	(0.0069)	(0.0054)	(0.0057)	(0.0054)
VT	0.0302***	0.0172***	0.0155**	0.0121**
	(0.0070)	(0.0055)	(0.0058)	(0.0055)
AZ	0.0231***	0.0129**	0.0097*	0.0082
	(0.0066)	(0.0063)	(0.0056)	(0.0067)
MD	0.0311***	0.0224***	0.0153**	0.0149***
	(0.0070)	(0.0047)	(0.0057)	(0.0053)
WA	0.0322***	0.0141**	0.0159***	0.0081
	(0.0069)	(0.0061)	(0.0050)	(0.0061)
NJ	0.0257***	0.0193***	0.0146**	0.0118**
	(0.0073)	(0.0047)	(0.0056)	(0.0055)
RI	0.0296***	0.0170***	0.0160***	0.0129**
	(0.0070)	(0.0054)	(0.0058)	(0.0055)
NY	0.0350***	0.0203***	0.0204***	0.0173***
	(0.0084)	(0.0069)	(0.0073)	(0.0060)
CO	0.0313***	0.0163***	0.0155**	0.0104**
	(0.0079)	(0.0056)	(0.0063)	(0.0051)
NM	0.0295***	0.0174***	0.0154**	0.0122**
	(0.0070)	(0.0056)	(0.0059)	(0.0055)

Note: This table presents Sun–Abraham (2021) interaction-weighted ATT estimates after dropping the indicated treated state from the analytic sample. The first row (None) reproduces the full-sample baseline from Table 3. Never-treated control states remain in every specification. Minnesota contributes no post-treatment observations to the static ATT because its lagged coded treatment year (2025) falls outside the panel and is therefore not listed as a treated-state omission. All other specification elements follow Table 3. Standard errors are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.